

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent
Kind Code
Date of Patent
Inventor(s)

12402816
B2
September 02, 2025
Chen; Jesse et al.

Medical monitoring device for harmonizing physiological measurements

Abstract

Systems, methods, apparatuses, and medical devices for harmonizing data from a plurality of non-invasive sensors are described. A physiological parameter can be determined by harmonizing data between two or more different types of non-invasive physiological sensors interrogating the same or proximate measurement sites. Data from one or more first non-invasive sensors can be utilized to identify one or more variables that are useful in one or more calculations associated with data from one or more second non-invasive sensors. Data from one or more first non-invasive sensors can be utilized to calibrate one or more second non-invasive sensors. Non-invasive sensors can include, but are not limited to, an optical coherence tomography (OCT) sensor, a bio-impedance sensor, a tissue dielectric constant sensor, a plethysmograph sensor, or a Raman spectrometer.

Inventors: Chen; Jesse (Foothill Ranch, CA), Merritt; Sean (Lake Forest, CA), Dalvi; Cristiano (Lake Forest, CA), Lesmana; Ferdyan (Irvine, CA), Vo; Hung The (Fountain Valley, CA), Pauley; Kevin Hughes (Lake Forest, CA), Poeze; Jeroen (Rancho Santa Margarita, CA), Long; Ruiqi (Irvine, CA), Monfre; Stephen L. (Rancho Santa Margarita, CA)

Applicant: Willow Laboratories, Inc. (Irvine, CA)

Family ID: 63174396

Assignee: Willow Laboratories, Inc. (Irvine, CA)

Appl. No.: 18/643195

Filed: April 23, 2024

Prior Publication Data

Document Identifier
US 20240324910 A1

Publication Date
Oct. 03, 2024

Related U.S. Application Data

continuation parent-doc US 17324396 20210519 US 11992311 child-doc US 18643195
continuation parent-doc US 16034743 20180713 US 11026604 20210608 child-doc US 17324396
us-provisional-application US 62667983 20180507
us-provisional-application US 62532273 20170713

Publication Classification

Int. Cl.: **A61B5/02** (20060101); **A61B5/00** (20060101); **A61B5/0205** (20060101); **A61B5/145** (20060101); A61B5/026 (20060101); A61B5/0295 (20060101); A61B5/0531 (20210101); A61B5/107 (20060101); A61B5/1455 (20060101)

U.S. Cl.:

CPC **A61B5/14532** (20130101); **A61B5/0075** (20130101); **A61B5/0205** (20130101); **A61B5/7203** (20130101); **A61B5/7278** (20130101); A61B5/0066 (20130101); A61B5/0261 (20130101); A61B5/0295 (20130101); A61B5/0531 (20130101); A61B5/1075 (20130101); A61B5/1079 (20130101); A61B5/14551 (20130101); A61B5/442 (20130101); A61B5/443 (20130101); A61B2560/02 (20130101); A61B2576/02 (20130101)

Field of Classification Search

CPC: A61B (5/14532); A61B (5/0075); A61B (5/0205); A61B (5/7203); A61B (5/7278); A61B (5/0066); A61B (5/0261); A61B (5/0295); A61B (5/0531); A61B (5/1075); A61B (5/1079); A61B (5/14551); A61B (5/442); A61B (5/443); A61B (2560/0223); A61B (2576/02)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4960128	12/1989	Gordon et al.	N/A	N/A
4964408	12/1989	Hink et al.	N/A	N/A
5041187	12/1990	Hink et al.	N/A	N/A
5069213	12/1990	Hink et al.	N/A	N/A
5163438	12/1991	Gordon et al.	N/A	N/A
5319355	12/1993	Russek	N/A	N/A
5337744	12/1993	Branigan	N/A	N/A
5341805	12/1993	Stavridi et al.	N/A	N/A
5377676	12/1994	Vari et al.	N/A	N/A
5431170	12/1994	Mathews	N/A	N/A
5436499	12/1994	Namavar et al.	N/A	N/A
5452717	12/1994	Branigan et al.	N/A	N/A
5456252	12/1994	Vari et al.	N/A	N/A
5479934	12/1995	Imran	N/A	N/A
5482036	12/1995	Diab et al.	N/A	N/A

5490505	12/1995	Diab et al.	N/A	N/A
5494043	12/1995	O'Sullivan et al.	N/A	N/A
5533511	12/1995	Kaspari et al.	N/A	N/A
5534851	12/1995	Russek	N/A	N/A
5590649	12/1996	Caro et al.	N/A	N/A
5602924	12/1996	Durand et al.	N/A	N/A
5632272	12/1996	Diab et al.	N/A	N/A
5638816	12/1996	Kiani- Azarbayjany et al.	N/A	N/A
5638818	12/1996	Diab et al.	N/A	N/A
5645440	12/1996	Tobler et al.	N/A	N/A
5671914	12/1996	Kalkhoran et al.	N/A	N/A
5685299	12/1996	Diab et al.	N/A	N/A
5726440	12/1997	Kalkhoran et al.	N/A	N/A
D393830	12/1997	Tobler et al.	N/A	N/A
5743262	12/1997	Lepper, Jr. et al.	N/A	N/A
5747806	12/1997	Khalil et al.	N/A	N/A
5750994	12/1997	Schlager	N/A	N/A
5758644	12/1997	Diab et al.	N/A	N/A
5760910	12/1997	Lepper, Jr. et al.	N/A	N/A
5769785	12/1997	Diab et al.	N/A	N/A
5782757	12/1997	Diab et al.	N/A	N/A
5785659	12/1997	Caro et al.	N/A	N/A
5791347	12/1997	Flaherty et al.	N/A	N/A
5810734	12/1997	Caro et al.	N/A	N/A
5823950	12/1997	Diab et al.	N/A	N/A
5830131	12/1997	Caro et al.	N/A	N/A
5833618	12/1997	Caro et al.	N/A	N/A
5860919	12/1998	Kiani- Azarbayjany et al.	N/A	N/A
5890929	12/1998	Mills et al.	N/A	N/A
5904654	12/1998	Wohltmann et al.	N/A	N/A
5919134	12/1998	Diab	N/A	N/A
5934925	12/1998	Tobler et al.	N/A	N/A
5940182	12/1998	Lepper, Jr. et al.	N/A	N/A
5987343	12/1998	Kinast	N/A	N/A
5995855	12/1998	Kiani et al.	N/A	N/A
5997343	12/1998	Mills et al.	N/A	N/A
6002952	12/1998	Diab et al.	N/A	N/A
6010937	12/1999	Karam et al.	N/A	N/A
6011986	12/1999	Diab et al.	N/A	N/A
6027452	12/1999	Flaherty et al.	N/A	N/A
6036642	12/1999	Diab et al.	N/A	N/A
6040578	12/1999	Malin et al.	N/A	N/A
6045509	12/1999	Caro et al.	N/A	N/A
6066204	12/1999	Haven	N/A	N/A
6067462	12/1999	Diab et al.	N/A	N/A
6070093	12/1999	Oosta et al.	N/A	N/A
6081735	12/1999	Diab et al.	N/A	N/A
6088607	12/1999	Diab et al.	N/A	N/A

6110522	12/1999	Lepper, Jr. et al.	N/A	N/A
6115673	12/1999	Malin et al.	N/A	N/A
6124597	12/1999	Shehada et al.	N/A	N/A
6128521	12/1999	Marro et al.	N/A	N/A
6129675	12/1999	Jay	N/A	N/A
6144868	12/1999	Parker	N/A	N/A
6151516	12/1999	Kiani- Azarbayjany et al.	N/A	N/A
6152754	12/1999	Gerhardt et al.	N/A	N/A
6157850	12/1999	Diab et al.	N/A	N/A
6165005	12/1999	Mills et al.	N/A	N/A
6184521	12/2000	Coffin, IV et al.	N/A	N/A
6206830	12/2000	Diab et al.	N/A	N/A
6229856	12/2000	Diab et al.	N/A	N/A
6232609	12/2000	Snyder et al.	N/A	N/A
6236872	12/2000	Diab et al.	N/A	N/A
6241683	12/2000	Macklem et al.	N/A	N/A
6253097	12/2000	Aronow et al.	N/A	N/A
6255708	12/2000	Sudharsanan et al.	N/A	N/A
6256523	12/2000	Diab et al.	N/A	N/A
6263222	12/2000	Diab et al.	N/A	N/A
6278522	12/2000	Lepper, Jr. et al.	N/A	N/A
6280213	12/2000	Tobler et al.	N/A	N/A
6280381	12/2000	Malin et al.	N/A	N/A
6285896	12/2000	Tobler et al.	N/A	N/A
6301493	12/2000	Marro et al.	N/A	N/A
6308089	12/2000	von der Ruhr et al.	N/A	N/A
6317627	12/2000	Ennen et al.	N/A	N/A
6321100	12/2000	Parker	N/A	N/A
6325761	12/2000	Jay	N/A	N/A
6334065	12/2000	Al-Ali et al.	N/A	N/A
6343224	12/2001	Parker	N/A	N/A
6349228	12/2001	Kiani et al.	N/A	N/A
6360114	12/2001	Diab et al.	N/A	N/A
6368283	12/2001	Xu et al.	N/A	N/A
6371921	12/2001	Caro et al.	N/A	N/A
6377829	12/2001	Al-Ali	N/A	N/A
6388240	12/2001	Schulz et al.	N/A	N/A
6397091	12/2001	Diab et al.	N/A	N/A
6411373	12/2001	Garside et al.	N/A	N/A
6415167	12/2001	Blank et al.	N/A	N/A
6430437	12/2001	Marro	N/A	N/A
6430525	12/2001	Weber et al.	N/A	N/A
6463311	12/2001	Diab	N/A	N/A
6470199	12/2001	Kopotic et al.	N/A	N/A
6487429	12/2001	Hockersmith et al.	N/A	N/A
6501975	12/2001	Diab et al.	N/A	N/A
6505059	12/2002	Kollias et al.	N/A	N/A
6515273	12/2002	Al-Ali	N/A	N/A

6519487	12/2002	Parker	N/A	N/A
6525386	12/2002	Mills et al.	N/A	N/A
6526300	12/2002	Kiani et al.	N/A	N/A
6534012	12/2002	Hazen et al.	N/A	N/A
6541756	12/2002	Schulz et al.	N/A	N/A
6542764	12/2002	Al-Ali et al.	N/A	N/A
6580086	12/2002	Schulz et al.	N/A	N/A
6584336	12/2002	Ali et al.	N/A	N/A
6587196	12/2002	Stippick et al.	N/A	N/A
6587199	12/2002	Luu	N/A	N/A
6597932	12/2002	Tian et al.	N/A	N/A
6597933	12/2002	Kiani et al.	N/A	N/A
6606511	12/2002	Ali et al.	N/A	N/A
6632181	12/2002	Flaherty et al.	N/A	N/A
6635559	12/2002	Greenwald et al.	N/A	N/A
6639668	12/2002	Trepagnier	N/A	N/A
6640116	12/2002	Diab	N/A	N/A
6640117	12/2002	Makarewicz et al.	N/A	N/A
6643530	12/2002	Diab et al.	N/A	N/A
6650917	12/2002	Diab et al.	N/A	N/A
6654624	12/2002	Diab et al.	N/A	N/A
6658276	12/2002	Kiani et al.	N/A	N/A
6661161	12/2002	Lanzo et al.	N/A	N/A
6671531	12/2002	Al-Ali	N/A	N/A
6678543	12/2003	Diab et al.	N/A	N/A
6684090	12/2003	Ali et al.	N/A	N/A
6684091	12/2003	Parker	N/A	N/A
6697656	12/2003	Al-Ali	N/A	N/A
6697657	12/2003	Shehada et al.	N/A	N/A
6697658	12/2003	Al-Ali	N/A	N/A
RE38476	12/2003	Diab et al.	N/A	N/A
6699194	12/2003	Diab et al.	N/A	N/A
6714804	12/2003	Al-Ali et al.	N/A	N/A
RE38492	12/2003	Diab et al.	N/A	N/A
6721582	12/2003	Trepagnier et al.	N/A	N/A
6721585	12/2003	Parker	N/A	N/A
6725075	12/2003	Al-Ali	N/A	N/A
6728560	12/2003	Kollias et al.	N/A	N/A
6735459	12/2003	Parker	N/A	N/A
6738652	12/2003	Mattu et al.	N/A	N/A
6745060	12/2003	Diab et al.	N/A	N/A
6760607	12/2003	Al-Ali	N/A	N/A
6770028	12/2003	Ali et al.	N/A	N/A
6771994	12/2003	Kiani et al.	N/A	N/A
6788965	12/2003	Ruchti et al.	N/A	N/A
6792300	12/2003	Diab et al.	N/A	N/A
6813511	12/2003	Diab et al.	N/A	N/A
6816241	12/2003	Grubisic	N/A	N/A
6816741	12/2003	Diab	N/A	N/A
6822564	12/2003	Al-Ali	N/A	N/A

6826419	12/2003	Diab et al.	N/A	N/A
6830711	12/2003	Mills et al.	N/A	N/A
6850787	12/2004	Weber et al.	N/A	N/A
6850788	12/2004	Al-Ali	N/A	N/A
6852083	12/2004	Caro et al.	N/A	N/A
6861639	12/2004	Al-Ali	N/A	N/A
6868285	12/2004	Muller-Dethlefs	N/A	N/A
6876931	12/2004	Lorenz et al.	N/A	N/A
6898452	12/2004	Al-Ali et al.	N/A	N/A
6920345	12/2004	Al-Ali et al.	N/A	N/A
6931268	12/2004	Kiani- Azarbayjany et al.	N/A	N/A
6934570	12/2004	Kiani et al.	N/A	N/A
6939305	12/2004	Flaherty et al.	N/A	N/A
6943348	12/2004	Coffin, IV	N/A	N/A
6950687	12/2004	Al-Ali	N/A	N/A
6956649	12/2004	Acosta et al.	N/A	N/A
6961598	12/2004	Diab	N/A	N/A
6970792	12/2004	Diab	N/A	N/A
6979812	12/2004	Al-Ali	N/A	N/A
6985764	12/2005	Mason et al.	N/A	N/A
6990364	12/2005	Ruchti et al.	N/A	N/A
6993371	12/2005	Kiani et al.	N/A	N/A
6996427	12/2005	Ali et al.	N/A	N/A
6998247	12/2005	Monfre et al.	N/A	N/A
6999904	12/2005	Weber et al.	N/A	N/A
7003338	12/2005	Weber et al.	N/A	N/A
7003339	12/2005	Diab et al.	N/A	N/A
7015451	12/2005	Dalke et al.	N/A	N/A
7024233	12/2005	Ali et al.	N/A	N/A
7027849	12/2005	Al-Ali	N/A	N/A
7030749	12/2005	Al-Ali	N/A	N/A
7039449	12/2005	Al-Ali	N/A	N/A
7041060	12/2005	Flaherty et al.	N/A	N/A
7044918	12/2005	Diab	N/A	N/A
7048687	12/2005	Reuss et al.	N/A	N/A
7067893	12/2005	Mills et al.	N/A	N/A
D526719	12/2005	Richie, Jr. et al.	N/A	N/A
7096052	12/2005	Mason et al.	N/A	N/A
7096054	12/2005	Abdul-Hafiz et al.	N/A	N/A
D529616	12/2005	Deros et al.	N/A	N/A
7132641	12/2005	Schulz et al.	N/A	N/A
7133710	12/2005	Acosta et al.	N/A	N/A
7142901	12/2005	Kiani et al.	N/A	N/A
7149561	12/2005	Diab	N/A	N/A
7186966	12/2006	Al-Ali	N/A	N/A
7190261	12/2006	Al-Ali	N/A	N/A
7215984	12/2006	Diab et al.	N/A	N/A
7215986	12/2006	Diab et al.	N/A	N/A
7221971	12/2006	Diab et al.	N/A	N/A

7225006	12/2006	Al-Ali et al.	N/A	N/A
7225007	12/2006	Al-Ali et al.	N/A	N/A
RE39672	12/2006	Shehada et al.	N/A	N/A
7239905	12/2006	Kiani- Azarbayjany et al.	N/A	N/A
7245953	12/2006	Parker	N/A	N/A
7254429	12/2006	Schurman et al.	N/A	N/A
7254430	12/2006	Cho et al.	N/A	N/A
7254431	12/2006	Al-Ali et al.	N/A	N/A
7254433	12/2006	Diab et al.	N/A	N/A
7254434	12/2006	Schulz et al.	N/A	N/A
7272425	12/2006	Al-Ali	N/A	N/A
7274955	12/2006	Kiani et al.	N/A	N/A
D554263	12/2006	Al-Ali et al.	N/A	N/A
7280858	12/2006	Al-Ali et al.	N/A	N/A
7289835	12/2006	Mansfield et al.	N/A	N/A
7292883	12/2006	De Felice et al.	N/A	N/A
7295866	12/2006	Al-Ali	N/A	N/A
7328053	12/2007	Diab et al.	N/A	N/A
7332784	12/2007	Mills et al.	N/A	N/A
7340287	12/2007	Mason et al.	N/A	N/A
7341559	12/2007	Schulz et al.	N/A	N/A
7343186	12/2007	Lamego et al.	N/A	N/A
D566282	12/2007	Al-Ali et al.	N/A	N/A
7355512	12/2007	Al-Ali	N/A	N/A
7356365	12/2007	Schurman	N/A	N/A
7371981	12/2007	Abdul-Hafiz	N/A	N/A
7373193	12/2007	Al-Ali et al.	N/A	N/A
7373194	12/2007	Weber et al.	N/A	N/A
7376453	12/2007	Diab et al.	N/A	N/A
7377794	12/2007	Al-Ali et al.	N/A	N/A
7377899	12/2007	Weber et al.	N/A	N/A
7383070	12/2007	Diab et al.	N/A	N/A
7395158	12/2007	Monfre et al.	N/A	N/A
7415297	12/2007	Al-Ali et al.	N/A	N/A
7428432	12/2007	Ali et al.	N/A	N/A
7438683	12/2007	Al-Ali et al.	N/A	N/A
7440787	12/2007	Diab	N/A	N/A
7454240	12/2007	Diab et al.	N/A	N/A
7467002	12/2007	Weber et al.	N/A	N/A
7469157	12/2007	Diab et al.	N/A	N/A
7471969	12/2007	Diab et al.	N/A	N/A
7471971	12/2007	Diab et al.	N/A	N/A
7483729	12/2008	Al-Ali et al.	N/A	N/A
7483730	12/2008	Diab et al.	N/A	N/A
7489958	12/2008	Diab et al.	N/A	N/A
7496391	12/2008	Diab et al.	N/A	N/A
7496393	12/2008	Diab et al.	N/A	N/A
D587657	12/2008	Al-Ali et al.	N/A	N/A
7499741	12/2008	Diab et al.	N/A	N/A

7499835	12/2008	Weber et al.	N/A	N/A
7500950	12/2008	Al-Ali et al.	N/A	N/A
7509154	12/2008	Diab et al.	N/A	N/A
7509494	12/2008	Al-Ali	N/A	N/A
7510849	12/2008	Schurman et al.	N/A	N/A
7514725	12/2008	Wojtczuk et al.	N/A	N/A
7519406	12/2008	Blank et al.	N/A	N/A
7526328	12/2008	Diab et al.	N/A	N/A
D592507	12/2008	Wachman et al.	N/A	N/A
7530942	12/2008	Diab	N/A	N/A
7530949	12/2008	Al Ali et al.	N/A	N/A
7530955	12/2008	Diab et al.	N/A	N/A
7563110	12/2008	Al-Ali et al.	N/A	N/A
7593230	12/2008	Abul-Haj et al.	N/A	N/A
7596398	12/2008	Al-Ali et al.	N/A	N/A
7606608	12/2008	Blank et al.	N/A	N/A
7618375	12/2008	Flaherty et al.	N/A	N/A
7620674	12/2008	Ruchti et al.	N/A	N/A
D606659	12/2008	Kiani et al.	N/A	N/A
7629039	12/2008	Eckerbom et al.	N/A	N/A
7640140	12/2008	Ruchti et al.	N/A	N/A
7647083	12/2009	Al-Ali et al.	N/A	N/A
D609193	12/2009	Al-Ali et al.	N/A	N/A
D614305	12/2009	Al-Ali et al.	N/A	N/A
7697966	12/2009	Monfre et al.	N/A	N/A
7698105	12/2009	Ruchti et al.	N/A	N/A
RE41317	12/2009	Parker	N/A	N/A
RE41333	12/2009	Blank et al.	N/A	N/A
7729733	12/2009	Al-Ali et al.	N/A	N/A
7734320	12/2009	Al-Ali	N/A	N/A
7761127	12/2009	Al-Ali et al.	N/A	N/A
7761128	12/2009	Al-Ali et al.	N/A	N/A
7764982	12/2009	Dalke et al.	N/A	N/A
D621516	12/2009	Kiani et al.	N/A	N/A
7791155	12/2009	Diab	N/A	N/A
7801581	12/2009	Diab	N/A	N/A
7822452	12/2009	Schurman et al.	N/A	N/A
RE41912	12/2009	Parker	N/A	N/A
7844313	12/2009	Kiani et al.	N/A	N/A
7844314	12/2009	Al-Ali	N/A	N/A
7844315	12/2009	Al-Ali	N/A	N/A
7865222	12/2010	Weber et al.	N/A	N/A
7873497	12/2010	Weber et al.	N/A	N/A
7880606	12/2010	Al-Ali	N/A	N/A
7880626	12/2010	Al-Ali et al.	N/A	N/A
7891355	12/2010	Al-Ali et al.	N/A	N/A
7894868	12/2010	Al-Ali et al.	N/A	N/A
7899507	12/2010	Al-Ali et al.	N/A	N/A
7904132	12/2010	Weber et al.	N/A	N/A
7909772	12/2010	Popov et al.	N/A	N/A

7910875	12/2010	Al-Ali	N/A	N/A
7919713	12/2010	Al-Ali et al.	N/A	N/A
7937128	12/2010	Al-Ali	N/A	N/A
7937129	12/2010	Mason et al.	N/A	N/A
7937130	12/2010	Diab et al.	N/A	N/A
7941199	12/2010	Kiani	N/A	N/A
7951086	12/2010	Flaherty et al.	N/A	N/A
7957780	12/2010	Lamego et al.	N/A	N/A
7962188	12/2010	Kiani et al.	N/A	N/A
7962190	12/2010	Diab et al.	N/A	N/A
7976472	12/2010	Kiani	N/A	N/A
7988637	12/2010	Diab	N/A	N/A
7990382	12/2010	Kiani	N/A	N/A
7991446	12/2010	Ali et al.	N/A	N/A
8000761	12/2010	Al-Ali	N/A	N/A
8008088	12/2010	Bellott et al.	N/A	N/A
RE42753	12/2010	Kiani- Azarbayjany et al.	N/A	N/A
8019400	12/2010	Diab et al.	N/A	N/A
8028701	12/2010	Al-Ali et al.	N/A	N/A
8029765	12/2010	Bellott et al.	N/A	N/A
8036727	12/2010	Schurman et al.	N/A	N/A
8036728	12/2010	Diab et al.	N/A	N/A
8046040	12/2010	Ali et al.	N/A	N/A
8046041	12/2010	Diab et al.	N/A	N/A
8046042	12/2010	Diab et al.	N/A	N/A
8048040	12/2010	Kiani	N/A	N/A
8050728	12/2010	Al-Ali et al.	N/A	N/A
RE43169	12/2011	Parker	N/A	N/A
8118620	12/2011	Al-Ali et al.	N/A	N/A
8126528	12/2011	Diab et al.	N/A	N/A
8128572	12/2011	Diab et al.	N/A	N/A
8130105	12/2011	Al-Ali et al.	N/A	N/A
8145287	12/2011	Diab et al.	N/A	N/A
8150487	12/2011	Diab et al.	N/A	N/A
8175672	12/2011	Parker	N/A	N/A
8180420	12/2011	Diab et al.	N/A	N/A
8182443	12/2011	Kiani	N/A	N/A
8185180	12/2011	Diab et al.	N/A	N/A
8190223	12/2011	Al-Ali et al.	N/A	N/A
8190227	12/2011	Diab et al.	N/A	N/A
8203438	12/2011	Kiani et al.	N/A	N/A
8203704	12/2011	Merritt et al.	N/A	N/A
8204566	12/2011	Schurman et al.	N/A	N/A
8219172	12/2011	Schurman et al.	N/A	N/A
8224411	12/2011	Al-Ali et al.	N/A	N/A
8228181	12/2011	Al-Ali	N/A	N/A
8229532	12/2011	Davis	N/A	N/A
8229533	12/2011	Diab et al.	N/A	N/A
8233955	12/2011	Al-Ali et al.	N/A	N/A

8244325	12/2011	Al-Ali et al.	N/A	N/A
8255026	12/2011	Al-Ali	N/A	N/A
8255027	12/2011	Al-Ali et al.	N/A	N/A
8255028	12/2011	Al-Ali et al.	N/A	N/A
8260577	12/2011	Weber et al.	N/A	N/A
8265723	12/2011	McHale et al.	N/A	N/A
8274360	12/2011	Sampath et al.	N/A	N/A
8280473	12/2011	Al-Ali	N/A	N/A
8301217	12/2011	Al-Ali et al.	N/A	N/A
8306596	12/2011	Schurman et al.	N/A	N/A
8310336	12/2011	Muhsin et al.	N/A	N/A
8315683	12/2011	Al-Ali et al.	N/A	N/A
RE43860	12/2011	Parker	N/A	N/A
8337403	12/2011	Al-Ali et al.	N/A	N/A
8346330	12/2012	Lamego	N/A	N/A
8353842	12/2012	Al-Ali et al.	N/A	N/A
8355766	12/2012	MacNeish et al.	N/A	N/A
8359080	12/2012	Diab et al.	N/A	N/A
8364223	12/2012	Al-Ali et al.	N/A	N/A
8364226	12/2012	Diab et al.	N/A	N/A
8374665	12/2012	Lamego	N/A	N/A
8385995	12/2012	Al-Ali et al.	N/A	N/A
8385996	12/2012	Smith et al.	N/A	N/A
8388353	12/2012	Kiani et al.	N/A	N/A
8399822	12/2012	Al-Ali	N/A	N/A
8401602	12/2012	Kiani	N/A	N/A
8405608	12/2012	Al-Ali et al.	N/A	N/A
8414499	12/2012	Al-Ali et al.	N/A	N/A
8418524	12/2012	Al-Ali	N/A	N/A
8423106	12/2012	Lamego et al.	N/A	N/A
8428967	12/2012	Olsen et al.	N/A	N/A
8430817	12/2012	Al-Ali et al.	N/A	N/A
8437825	12/2012	Dalvi et al.	N/A	N/A
8455290	12/2012	Siskavich	N/A	N/A
8457703	12/2012	Al-Ali	N/A	N/A
8457707	12/2012	Kiani	N/A	N/A
8463349	12/2012	Diab et al.	N/A	N/A
8466286	12/2012	Bellott et al.	N/A	N/A
8471713	12/2012	Poeze et al.	N/A	N/A
8473020	12/2012	Kiani et al.	N/A	N/A
8483787	12/2012	Al-Ali et al.	N/A	N/A
8489364	12/2012	Weber et al.	N/A	N/A
8498684	12/2012	Weber et al.	N/A	N/A
8504128	12/2012	Blank et al.	N/A	N/A
8509867	12/2012	Workman et al.	N/A	N/A
8515509	12/2012	Bruinsma et al.	N/A	N/A
8523781	12/2012	Al-Ali	N/A	N/A
8529301	12/2012	Al-Ali et al.	N/A	N/A
8532727	12/2012	Ali et al.	N/A	N/A
8532728	12/2012	Diab et al.	N/A	N/A

D692145	12/2012	Al-Ali et al.	N/A	N/A
8547209	12/2012	Kiani et al.	N/A	N/A
8548548	12/2012	Ai-All	N/A	N/A
8548549	12/2012	Schurman et al.	N/A	N/A
8548550	12/2012	Al-Ali et al.	N/A	N/A
8560032	12/2012	Al-Ali et al.	N/A	N/A
8560034	12/2012	Diab et al.	N/A	N/A
8570167	12/2012	Al-Ali	N/A	N/A
8570503	12/2012	Vo et al.	N/A	N/A
8571617	12/2012	Reichgott et al.	N/A	N/A
8571618	12/2012	Lamego et al.	N/A	N/A
8571619	12/2012	Al-Ali et al.	N/A	N/A
8577431	12/2012	Lamego et al.	N/A	N/A
8581732	12/2012	Al-Ali et al.	N/A	N/A
8584345	12/2012	Al-Ali et al.	N/A	N/A
8588880	12/2012	Abdul-Hafiz et al.	N/A	N/A
8600467	12/2012	Al-Ali et al.	N/A	N/A
8606342	12/2012	Diab	N/A	N/A
8626255	12/2013	Al-Ali et al.	N/A	N/A
8630691	12/2013	Lamego et al.	N/A	N/A
8634889	12/2013	Al-Ali et al.	N/A	N/A
8641631	12/2013	Sierra et al.	N/A	N/A
8652060	12/2013	Al-Ali	N/A	N/A
8663107	12/2013	Kiani	N/A	N/A
8666468	12/2013	Al-Ali	N/A	N/A
8667967	12/2013	Al-Ali et al.	N/A	N/A
8670811	12/2013	O'Reilly	N/A	N/A
8670814	12/2013	Diab et al.	N/A	N/A
8676286	12/2013	Weber et al.	N/A	N/A
8682407	12/2013	Al-Ali	N/A	N/A
RE44823	12/2013	Parker	N/A	N/A
RE44875	12/2013	Kiani et al.	N/A	N/A
8688183	12/2013	Bruinsma et al.	N/A	N/A
8690799	12/2013	Telfort et al.	N/A	N/A
8700112	12/2013	Kiani	N/A	N/A
8702627	12/2013	Telfort et al.	N/A	N/A
8706179	12/2013	Parker	N/A	N/A
8712494	12/2013	MacNeish, III et al.	N/A	N/A
8715206	12/2013	Telfort et al.	N/A	N/A
8718735	12/2013	Lamego et al.	N/A	N/A
8718737	12/2013	Diab et al.	N/A	N/A
8718738	12/2013	Blank et al.	N/A	N/A
8720249	12/2013	Al-Ali	N/A	N/A
8721541	12/2013	Al-Ali et al.	N/A	N/A
8721542	12/2013	Al-Ali et al.	N/A	N/A
8723677	12/2013	Kiani	N/A	N/A
8740792	12/2013	Kiani et al.	N/A	N/A
8754776	12/2013	Poeze et al.	N/A	N/A
8755535	12/2013	Telfort et al.	N/A	N/A

8755856	12/2013	Diab et al.	N/A	N/A
8755872	12/2013	Marinow	N/A	N/A
8761850	12/2013	Lamego	N/A	N/A
8764671	12/2013	Kiani	N/A	N/A
8768423	12/2013	Shakespeare et al.	N/A	N/A
8771204	12/2013	Telfort et al.	N/A	N/A
8777634	12/2013	Kiani et al.	N/A	N/A
8781543	12/2013	Diab et al.	N/A	N/A
8781544	12/2013	Al-Ali et al.	N/A	N/A
8781549	12/2013	Al-Ali et al.	N/A	N/A
8788003	12/2013	Schurman et al.	N/A	N/A
8790268	12/2013	Al-Ali	N/A	N/A
8801613	12/2013	Al-Ali et al.	N/A	N/A
8821397	12/2013	Al-Ali et al.	N/A	N/A
8821415	12/2013	Al-Ali et al.	N/A	N/A
8830449	12/2013	Lamego et al.	N/A	N/A
8831700	12/2013	Schurman et al.	N/A	N/A
8840549	12/2013	Al-Ali et al.	N/A	N/A
8847740	12/2013	Kiani et al.	N/A	N/A
8849365	12/2013	Smith et al.	N/A	N/A
8852094	12/2013	Al-Ali et al.	N/A	N/A
8852994	12/2013	Wojtczuk et al.	N/A	N/A
8868147	12/2013	Stippick et al.	N/A	N/A
8868150	12/2013	Al-Ali et al.	N/A	N/A
8870792	12/2013	Al-Ali et al.	N/A	N/A
8886271	12/2013	Kiani et al.	N/A	N/A
8888539	12/2013	Al-Ali et al.	N/A	N/A
8888708	12/2013	Diab et al.	N/A	N/A
8892180	12/2013	Weber et al.	N/A	N/A
8897847	12/2013	Al-Ali	N/A	N/A
8909310	12/2013	Lamego et al.	N/A	N/A
8911377	12/2013	Ai-Ali	N/A	N/A
8912909	12/2013	Al-Ali et al.	N/A	N/A
8920317	12/2013	Al-Ali et al.	N/A	N/A
8921699	12/2013	Al-Ali et al.	N/A	N/A
8922382	12/2013	Al-Ali et al.	N/A	N/A
8929964	12/2014	Al-Ali et al.	N/A	N/A
8942777	12/2014	Diab et al.	N/A	N/A
8948834	12/2014	Diab et al.	N/A	N/A
8948835	12/2014	Diab	N/A	N/A
8965471	12/2014	Lamego	N/A	N/A
8983564	12/2014	Al-Ali	N/A	N/A
8989831	12/2014	Al-Ali et al.	N/A	N/A
8996085	12/2014	Kiani et al.	N/A	N/A
8998809	12/2014	Kiani	N/A	N/A
9028429	12/2014	Telfort et al.	N/A	N/A
9037207	12/2014	Al-Ali et al.	N/A	N/A
9060721	12/2014	Reichgott et al.	N/A	N/A
9066666	12/2014	Kiani	N/A	N/A
9066680	12/2014	Al-Ali et al.	N/A	N/A

9072474	12/2014	Al-Ali et al.	N/A	N/A
9078560	12/2014	Schurman et al.	N/A	N/A
9084569	12/2014	Weber et al.	N/A	N/A
9095316	12/2014	Welch et al.	N/A	N/A
9106038	12/2014	Telfort et al.	N/A	N/A
9107625	12/2014	Telfort et al.	N/A	N/A
9107626	12/2014	Al-Ali et al.	N/A	N/A
9113831	12/2014	Al-Ali	N/A	N/A
9113832	12/2014	Al-Ali	N/A	N/A
9119595	12/2014	Lamego	N/A	N/A
9131881	12/2014	Diab et al.	N/A	N/A
9131882	12/2014	Al-Ali et al.	N/A	N/A
9131883	12/2014	Al-Ali	N/A	N/A
9131917	12/2014	Telfort et al.	N/A	N/A
9138180	12/2014	Coverston et al.	N/A	N/A
9138182	12/2014	Al-Ali et al.	N/A	N/A
9138192	12/2014	Weber et al.	N/A	N/A
9142117	12/2014	Muhsin et al.	N/A	N/A
9153112	12/2014	Kiani et al.	N/A	N/A
9153121	12/2014	Kiani et al.	N/A	N/A
9161696	12/2014	Al-Ali et al.	N/A	N/A
9161713	12/2014	Al-Ali et al.	N/A	N/A
9167995	12/2014	Lamego et al.	N/A	N/A
9176141	12/2014	Al-Ali et al.	N/A	N/A
9186102	12/2014	Bruinsma et al.	N/A	N/A
9192312	12/2014	Al-Ali	N/A	N/A
9192329	12/2014	Al-Ali	N/A	N/A
9192351	12/2014	Telfort et al.	N/A	N/A
9195385	12/2014	Al-Ali et al.	N/A	N/A
9211072	12/2014	Kiani	N/A	N/A
9211095	12/2014	Al-Ali	N/A	N/A
9218454	12/2014	Kiani et al.	N/A	N/A
9226696	12/2015	Kiani	N/A	N/A
9241662	12/2015	Al-Ali et al.	N/A	N/A
9245668	12/2015	Vo et al.	N/A	N/A
9259185	12/2015	Abdul-Hafiz et al.	N/A	N/A
9267572	12/2015	Barker et al.	N/A	N/A
9277880	12/2015	Poeze et al.	N/A	N/A
9289167	12/2015	Diab et al.	N/A	N/A
9295421	12/2015	Kiani et al.	N/A	N/A
9307928	12/2015	Al-Ali et al.	N/A	N/A
9323894	12/2015	Kiani	N/A	N/A
D755392	12/2015	Hwang et al.	N/A	N/A
9326712	12/2015	Kiani	N/A	N/A
9333316	12/2015	Kiani	N/A	N/A
9339220	12/2015	Lamego et al.	N/A	N/A
9341565	12/2015	Lamego et al.	N/A	N/A
9351673	12/2015	Diab et al.	N/A	N/A
9351675	12/2015	Al-Ali et al.	N/A	N/A
9364181	12/2015	Kiani et al.	N/A	N/A

9368671	12/2015	Wojtczuk et al.	N/A	N/A
9370325	12/2015	Al-Ali et al.	N/A	N/A
9370326	12/2015	McHale et al.	N/A	N/A
9370335	12/2015	Al-Ali et al.	N/A	N/A
9375185	12/2015	Ali et al.	N/A	N/A
9386953	12/2015	Ai-Ai	N/A	N/A
9386961	12/2015	Al-Ali et al.	N/A	N/A
9392945	12/2015	Al-Ali et al.	N/A	N/A
9397448	12/2015	Al-Ali et al.	N/A	N/A
9408542	12/2015	Kinast et al.	N/A	N/A
9436645	12/2015	Al-Ali et al.	N/A	N/A
9445759	12/2015	Lamego et al.	N/A	N/A
9466919	12/2015	Kiani et al.	N/A	N/A
9474474	12/2015	Lamego et al.	N/A	N/A
9480422	12/2015	Al-Ali	N/A	N/A
9480435	12/2015	Olsen	N/A	N/A
9492110	12/2015	Al-Ali et al.	N/A	N/A
9510779	12/2015	Poeze et al.	N/A	N/A
9517024	12/2015	Kiani et al.	N/A	N/A
9532722	12/2016	Lamego et al.	N/A	N/A
9538949	12/2016	Al-Ali et al.	N/A	N/A
9538980	12/2016	Telfort et al.	N/A	N/A
9549696	12/2016	Lamego et al.	N/A	N/A
9554737	12/2016	Schurman et al.	N/A	N/A
9560996	12/2016	Kiani	N/A	N/A
9560998	12/2016	Al-Ali et al.	N/A	N/A
9566019	12/2016	Al-Ali et al.	N/A	N/A
9579039	12/2016	Jansen et al.	N/A	N/A
9591975	12/2016	Dalvi et al.	N/A	N/A
9622692	12/2016	Lamego et al.	N/A	N/A
9622693	12/2016	Diab	N/A	N/A
D788312	12/2016	Al-Ali et al.	N/A	N/A
9636055	12/2016	Al Ali et al.	N/A	N/A
9636056	12/2016	Al-Ali	N/A	N/A
9649054	12/2016	Lamego et al.	N/A	N/A
9662052	12/2016	Al-Ali et al.	N/A	N/A
9668679	12/2016	Schurman et al.	N/A	N/A
9668680	12/2016	Bruinsma et al.	N/A	N/A
9668703	12/2016	Al-Ali	N/A	N/A
9675286	12/2016	Diab	N/A	N/A
9687160	12/2016	Kiani	N/A	N/A
9693719	12/2016	Al-Ali et al.	N/A	N/A
9693737	12/2016	Al-Ali	N/A	N/A
9697928	12/2016	Al-Ali et al.	N/A	N/A
9717425	12/2016	Kiani et al.	N/A	N/A
9717458	12/2016	Lamego et al.	N/A	N/A
9724016	12/2016	Al-Ali et al.	N/A	N/A
9724024	12/2016	Al-Ali	N/A	N/A
9724025	12/2016	Kiani et al.	N/A	N/A
9730640	12/2016	Diab et al.	N/A	N/A

9743887	12/2016	Al-Ali et al.	N/A	N/A
9749232	12/2016	Sampath et al.	N/A	N/A
9750442	12/2016	Olsen	N/A	N/A
9750443	12/2016	Smith et al.	N/A	N/A
9750461	12/2016	Telfort	N/A	N/A
9759714	12/2016	Bordelon et al.	N/A	N/A
9775545	12/2016	Al-Ali et al.	N/A	N/A
9775546	12/2016	Diab et al.	N/A	N/A
9775570	12/2016	Al-Ali	N/A	N/A
9778079	12/2016	Al-Ali et al.	N/A	N/A
9782077	12/2016	Lamego et al.	N/A	N/A
9782110	12/2016	Kiani	N/A	N/A
9787568	12/2016	Lamego et al.	N/A	N/A
9788735	12/2016	Al-Ali	N/A	N/A
9788768	12/2016	Al-Ali et al.	N/A	N/A
9795300	12/2016	Al-Ali	N/A	N/A
9795310	12/2016	Al-Ali	N/A	N/A
9795358	12/2016	Telfort et al.	N/A	N/A
9795739	12/2016	Al-Ali et al.	N/A	N/A
9801556	12/2016	Kiani	N/A	N/A
9801588	12/2016	Weber et al.	N/A	N/A
9808188	12/2016	Perea et al.	N/A	N/A
9814418	12/2016	Weber et al.	N/A	N/A
9820691	12/2016	Kiani	N/A	N/A
9833152	12/2016	Kiani et al.	N/A	N/A
9833180	12/2016	Shakespeare et al.	N/A	N/A
9839379	12/2016	Al-Ali et al.	N/A	N/A
9839381	12/2016	Weber et al.	N/A	N/A
9847002	12/2016	Kiani et al.	N/A	N/A
9847749	12/2016	Kiani et al.	N/A	N/A
9848800	12/2016	Lee et al.	N/A	N/A
9848806	12/2016	Al-Ali	N/A	N/A
9848807	12/2016	Lamego	N/A	N/A
9861298	12/2017	Eckerbom et al.	N/A	N/A
9861304	12/2017	Al-Ali et al.	N/A	N/A
9861305	12/2017	Weber et al.	N/A	N/A
9867578	12/2017	Al-Ali et al.	N/A	N/A
9872623	12/2017	Al-Ali	N/A	N/A
9876320	12/2017	Coverston et al.	N/A	N/A
9877650	12/2017	Muhsin et al.	N/A	N/A
9877686	12/2017	Al-Ali et al.	N/A	N/A
9891079	12/2017	Dalvi	N/A	N/A
9895107	12/2017	Al-Ali et al.	N/A	N/A
9924893	12/2017	Schurman et al.	N/A	N/A
9924897	12/2017	Abdul-Hafiz	N/A	N/A
9936917	12/2017	Poeze et al.	N/A	N/A
9955937	12/2017	Telfort	N/A	N/A
9965946	12/2017	Al-Ali et al.	N/A	N/A
D820865	12/2017	Muhsin et al.	N/A	N/A
9986952	12/2017	Dalvi et al.	N/A	N/A

D822215	12/2017	Al-Ali et al.	N/A	N/A
D822216	12/2017	Barker et al.	N/A	N/A
10010276	12/2017	Al-Ali et al.	N/A	N/A
10086138	12/2017	Novak, Jr.	N/A	N/A
10111591	12/2017	Dyell et al.	N/A	N/A
D833624	12/2017	DeJong et al.	N/A	N/A
10123729	12/2017	Dyell et al.	N/A	N/A
D835282	12/2017	Barker et al.	N/A	N/A
D835283	12/2017	Barker et al.	N/A	N/A
D835284	12/2017	Barker et al.	N/A	N/A
D835285	12/2017	Barker et al.	N/A	N/A
10149616	12/2017	Al-Ali et al.	N/A	N/A
10154815	12/2017	Al-Ali et al.	N/A	N/A
10159412	12/2017	Lamego et al.	N/A	N/A
10188348	12/2018	Al-Ali et al.	N/A	N/A
RE47218	12/2018	Al-Ali	N/A	N/A
RE47244	12/2018	Kiani et al.	N/A	N/A
RE47249	12/2018	Kiani et al.	N/A	N/A
10205291	12/2018	Scruggs et al.	N/A	N/A
10226187	12/2018	Al-Ali et al.	N/A	N/A
10231657	12/2018	Al-Ali et al.	N/A	N/A
10231670	12/2018	Blank et al.	N/A	N/A
RE47353	12/2018	Kiani et al.	N/A	N/A
10279247	12/2018	Kiani	N/A	N/A
10292664	12/2018	Al-Ali	N/A	N/A
10299720	12/2018	Brown et al.	N/A	N/A
10327337	12/2018	Schmidt et al.	N/A	N/A
10327713	12/2018	Barker et al.	N/A	N/A
10332630	12/2018	Al-Ali	N/A	N/A
10383520	12/2018	Wojtczuk et al.	N/A	N/A
10383527	12/2018	Al-Ali	N/A	N/A
10388120	12/2018	Muhsin et al.	N/A	N/A
D864120	12/2018	Forrest et al.	N/A	N/A
10441181	12/2018	Telfort et al.	N/A	N/A
10441196	12/2018	Eckerbom et al.	N/A	N/A
10448844	12/2018	Al-Ali et al.	N/A	N/A
10448871	12/2018	Al-Ali et al.	N/A	N/A
10456038	12/2018	Lamego et al.	N/A	N/A
10463340	12/2018	Telfort et al.	N/A	N/A
10471159	12/2018	Lapotko et al.	N/A	N/A
10505311	12/2018	Al-Ali et al.	N/A	N/A
10524738	12/2019	Olsen	N/A	N/A
10532174	12/2019	Al-Ali	N/A	N/A
10537285	12/2019	Shreim et al.	N/A	N/A
10542903	12/2019	Al-Ali et al.	N/A	N/A
10555678	12/2019	Dalvi et al.	N/A	N/A
10568553	12/2019	O'Neil et al.	N/A	N/A
RE47882	12/2019	Al-Ali	N/A	N/A
10608817	12/2019	Haider et al.	N/A	N/A
D880477	12/2019	Forrest et al.	N/A	N/A

10617302	12/2019	Al-Ali et al.	N/A	N/A
10617335	12/2019	Al-Ali et al.	N/A	N/A
10637181	12/2019	Al-Ali et al.	N/A	N/A
D886849	12/2019	Muhsin et al.	N/A	N/A
D887548	12/2019	Abdul-Hafiz et al.	N/A	N/A
D887549	12/2019	Abdul-Hafiz et al.	N/A	N/A
10667764	12/2019	Ahmed et al.	N/A	N/A
D890708	12/2019	Forrest et al.	N/A	N/A
10721785	12/2019	Al-Ali	N/A	N/A
10736518	12/2019	Al-Ali et al.	N/A	N/A
10750984	12/2019	Pauley et al.	N/A	N/A
D897098	12/2019	Al-Ali	N/A	N/A
10779098	12/2019	Iswanto et al.	N/A	N/A
10827961	12/2019	Iyengar et al.	N/A	N/A
10828007	12/2019	Telfort et al.	N/A	N/A
10832818	12/2019	Muhsin et al.	N/A	N/A
10835130	12/2019	Cho et al.	N/A	N/A
10849554	12/2019	Shreim et al.	N/A	N/A
10856750	12/2019	Indorf	N/A	N/A
D906970	12/2020	Forrest et al.	N/A	N/A
D908213	12/2020	Abdul-Hafiz et al.	N/A	N/A
10918281	12/2020	Al-Ali et al.	N/A	N/A
10932705	12/2020	Muhsin et al.	N/A	N/A
10932729	12/2020	Kiani et al.	N/A	N/A
10939878	12/2020	Kiani et al.	N/A	N/A
10956950	12/2020	Al-Ali et al.	N/A	N/A
10959651	12/2020	McKinney et al.	N/A	N/A
D916135	12/2020	Indorf et al.	N/A	N/A
D917046	12/2020	Abdul-Hafiz et al.	N/A	N/A
D917550	12/2020	Indorf et al.	N/A	N/A
D917564	12/2020	Indorf et al.	N/A	N/A
D917704	12/2020	Al-Ali et al.	N/A	N/A
10987066	12/2020	Chandran et al.	N/A	N/A
10991135	12/2020	Al-Ali et al.	N/A	N/A
D919094	12/2020	Al-Ali et al.	N/A	N/A
D919100	12/2020	Al-Ali et al.	N/A	N/A
11006867	12/2020	Al-Ali	N/A	N/A
D921202	12/2020	Al-Ali et al.	N/A	N/A
11024064	12/2020	Muhsin et al.	N/A	N/A
11026604	12/2020	Chen	N/A	A61B 5/0205
D925597	12/2020	Chandran et al.	N/A	N/A
D927699	12/2020	Al-Ali et al.	N/A	N/A
11076777	12/2020	Lee et al.	N/A	N/A
11114188	12/2020	Poeze et al.	N/A	N/A
D933232	12/2020	Al-Ali et al.	N/A	N/A
D933233	12/2020	Al-Ali et al.	N/A	N/A
D933234	12/2020	Al-Ali et al.	N/A	N/A
11145408	12/2020	Sampath et al.	N/A	N/A
11147518	12/2020	Al-Ali et al.	N/A	N/A

11185262	12/2020	Al-Ali et al.	N/A	N/A
11191484	12/2020	Kiani et al.	N/A	N/A
D946596	12/2021	Ahmed	N/A	N/A
D946597	12/2021	Ahmed	N/A	N/A
D946598	12/2021	Ahmed	N/A	N/A
D946617	12/2021	Ahmed	N/A	N/A
11272839	12/2021	Al-Ali et al.	N/A	N/A
11289199	12/2021	Al-Ali	N/A	N/A
RE49034	12/2021	Al-Ali	N/A	N/A
11298021	12/2021	Muhsin et al.	N/A	N/A
D950580	12/2021	Ahmed	N/A	N/A
D950599	12/2021	Ahmed	N/A	N/A
D950738	12/2021	Al-Ali et al.	N/A	N/A
D957648	12/2021	Al-Ali	N/A	N/A
11382567	12/2021	O'Brien et al.	N/A	N/A
11389093	12/2021	Triman et al.	N/A	N/A
11406286	12/2021	Al-Ali et al.	N/A	N/A
11417426	12/2021	Muhsin et al.	N/A	N/A
11439329	12/2021	Lamego	N/A	N/A
11445948	12/2021	Scruggs et al.	N/A	N/A
D965789	12/2021	Al-Ali et al.	N/A	N/A
D967433	12/2021	Al-Ali et al.	N/A	N/A
11464410	12/2021	Muhsin	N/A	N/A
11504058	12/2021	Sharma et al.	N/A	N/A
11504066	12/2021	Dalvi et al.	N/A	N/A
D971933	12/2021	Ahmed	N/A	N/A
D973072	12/2021	Ahmed	N/A	N/A
D973685	12/2021	Ahmed	N/A	N/A
D973686	12/2021	Ahmed	N/A	N/A
D974193	12/2022	Forrest et al.	N/A	N/A
D979516	12/2022	Al-Ali et al.	N/A	N/A
D980091	12/2022	Forrest et al.	N/A	N/A
11596363	12/2022	Lamego	N/A	N/A
11627919	12/2022	Kiani et al.	N/A	N/A
11637437	12/2022	Al-Ali et al.	N/A	N/A
D985498	12/2022	Al-Ali et al.	N/A	N/A
11653862	12/2022	Dalvi et al.	N/A	N/A
D989112	12/2022	Muhsin et al.	N/A	N/A
D989327	12/2022	Al-Ali et al.	N/A	N/A
11678829	12/2022	Al-Ali et al.	N/A	N/A
11679579	12/2022	Al-Ali	N/A	N/A
11684296	12/2022	Vo et al.	N/A	N/A
11692934	12/2022	Normand et al.	N/A	N/A
11701043	12/2022	Al-Ali et al.	N/A	N/A
D997365	12/2022	Hwang	N/A	N/A
11721105	12/2022	Ranasinghe et al.	N/A	N/A
11730379	12/2022	Ahmed et al.	N/A	N/A
D998625	12/2022	Indorf et al.	N/A	N/A
D998630	12/2022	Indorf et al.	N/A	N/A
D998631	12/2022	Indorf et al.	N/A	N/A

D999244	12/2022	Indorf et al.	N/A	N/A
D999245	12/2022	Indorf et al.	N/A	N/A
D999246	12/2022	Indorf et al.	N/A	N/A
11766198	12/2022	Pauley et al.	N/A	N/A
D1000975	12/2022	Al-Ali et al.	N/A	N/A
11803623	12/2022	Kiani et al.	N/A	N/A
11832940	12/2022	Diab et al.	N/A	N/A
D1013179	12/2023	Al-Ali et al.	N/A	N/A
11872156	12/2023	Telfort et al.	N/A	N/A
11879960	12/2023	Ranasinghe et al.	N/A	N/A
11883129	12/2023	Olsen	N/A	N/A
D1022729	12/2023	Forrest et al.	N/A	N/A
11951186	12/2023	Krishnamani et al.	N/A	N/A
11974833	12/2023	Forrest et al.	N/A	N/A
11986067	12/2023	Al-Ali et al.	N/A	N/A
11986289	12/2023	Dalvi et al.	N/A	N/A
11986305	12/2023	Al-Ali et al.	N/A	N/A
11992311	12/2023	Chen	N/A	A61B 5/0205
D1031729	12/2023	Forrest et al.	N/A	N/A
12004869	12/2023	Kiani et al.	N/A	N/A
12014328	12/2023	Wachman et al.	N/A	N/A
D1036293	12/2023	Al-Ali et al.	N/A	N/A
D1037462	12/2023	Al-Ali et al.	N/A	N/A
12029844	12/2023	Pauley et al.	N/A	N/A
12048534	12/2023	Vo et al.	N/A	N/A
12064217	12/2023	Ahmed et al.	N/A	N/A
12066426	12/2023	Lapotko et al.	N/A	N/A
D1041511	12/2023	Indorf et al.	N/A	N/A
D1042596	12/2023	DeJong et al.	N/A	N/A
D1042852	12/2023	Hwang	N/A	N/A
12076159	12/2023	Belur Nagaraj et al.	N/A	N/A
12082926	12/2023	Sharma et al.	N/A	N/A
D1044828	12/2023	Chandran et al.	N/A	N/A
D1048571	12/2023	Yu et al.	N/A	N/A
D1048908	12/2023	Al-Ali et al.	N/A	N/A
12106752	12/2023	Campbell et al.	N/A	N/A
12114974	12/2023	Al-Ali et al.	N/A	N/A
12126683	12/2023	Koo et al.	N/A	N/A
12127838	12/2023	Olsen et al.	N/A	N/A
12128213	12/2023	Kiani et al.	N/A	N/A
12131661	12/2023	Pauley et al.	N/A	N/A
D1050910	12/2023	Al-Ali et al.	N/A	N/A
12178572	12/2023	Pauley et al.	N/A	N/A
12178581	12/2023	Telfort et al.	N/A	N/A
12178852	12/2023	Kiani et al.	N/A	N/A
D1057159	12/2024	DeJong et al.	N/A	N/A
D1057160	12/2024	DeJong et al.	N/A	N/A
12198790	12/2024	Al-Ali	N/A	N/A

12200421	12/2024	Campbell et al.	N/A	N/A
12207901	12/2024	Lapotko et al.	N/A	N/A
D1060680	12/2024	Al-Ali et al.	N/A	N/A
D1061585	12/2024	Indorf	N/A	N/A
D1063893	12/2024	DeJong et al.	N/A	N/A
12220207	12/2024	Telfort et al.	N/A	N/A
12235941	12/2024	Kiani et al.	N/A	N/A
12236767	12/2024	Muhsin	N/A	N/A
D1066244	12/2024	Lim et al.	N/A	N/A
D1066672	12/2024	Al-Ali et al.	N/A	N/A
2001/0034477	12/2000	Mansfield et al.	N/A	N/A
2001/0039483	12/2000	Brand et al.	N/A	N/A
2002/0010401	12/2001	Bushmakina et al.	N/A	N/A
2002/0058864	12/2001	Mansfield et al.	N/A	N/A
2002/0133080	12/2001	Apruzzese et al.	N/A	N/A
2003/0013975	12/2002	Kiani	N/A	N/A
2003/0018243	12/2002	Gerhardt et al.	N/A	N/A
2003/0144582	12/2002	Cohen et al.	N/A	N/A
2003/0156288	12/2002	Barnum et al.	N/A	N/A
2003/0212312	12/2002	Coffin, IV et al.	N/A	N/A
2004/0106163	12/2003	Workman, Jr. et al.	N/A	N/A
2005/0055276	12/2004	Kiani et al.	N/A	N/A
2005/0234317	12/2004	Kiani	N/A	N/A
2006/0073719	12/2005	Kiani	N/A	N/A
2006/0161054	12/2005	Reuss et al.	N/A	N/A
2006/0189871	12/2005	Al-Ali et al.	N/A	N/A
2006/0200017	12/2005	Monfre et al.	N/A	N/A
2006/0211926	12/2005	Yu et al.	N/A	N/A
2007/0073116	12/2006	Kiani et al.	N/A	N/A
2007/0180140	12/2006	Welch et al.	N/A	N/A
2007/0244377	12/2006	Cozad et al.	N/A	N/A
2007/0282478	12/2006	Al-Ali et al.	N/A	N/A
2008/0064965	12/2007	Jay et al.	N/A	N/A
2008/0094228	12/2007	Welch et al.	N/A	N/A
2008/0103375	12/2007	Kiani	N/A	N/A
2008/0221418	12/2007	Al-Ali et al.	N/A	N/A
2009/0036759	12/2008	Ault et al.	N/A	N/A
2009/0093687	12/2008	Telfort et al.	N/A	N/A
2009/0095926	12/2008	MacNeish, III	N/A	N/A
2009/0247984	12/2008	Lamego et al.	N/A	N/A
2009/0275813	12/2008	Davis	N/A	N/A
2009/0275844	12/2008	Al-Ali	N/A	N/A
2010/0004518	12/2009	Vo et al.	N/A	N/A
2010/0030040	12/2009	Poeze et al.	N/A	N/A
2010/0099964	12/2009	O'Reilly et al.	N/A	N/A
2010/0234718	12/2009	Sampath et al.	N/A	N/A
2010/0270257	12/2009	Wachman et al.	N/A	N/A
2011/0028806	12/2010	Merritt et al.	N/A	N/A
2011/0028809	12/2010	Goodman	N/A	N/A

2011/0040197	12/2010	Welch et al.	N/A	N/A
2011/0082711	12/2010	Poeze et al.	N/A	N/A
2011/0087081	12/2010	Kiani et al.	N/A	N/A
2011/0105854	12/2010	Kiani et al.	N/A	N/A
2011/0118561	12/2010	Tari et al.	N/A	N/A
2011/0125060	12/2010	Telfort et al.	N/A	N/A
2011/0137297	12/2010	Kiani et al.	N/A	N/A
2011/0172498	12/2010	Olsen et al.	N/A	N/A
2011/0208015	12/2010	Welch et al.	N/A	N/A
2011/0230733	12/2010	Al-Ali	N/A	N/A
2012/0123231	12/2011	O'Reilly	N/A	N/A
2012/0165629	12/2011	Merritt et al.	N/A	N/A
2012/0188538	12/2011	Patil et al.	N/A	N/A
2012/0209082	12/2011	Al-Ali	N/A	N/A
2012/0209084	12/2011	Olsen et al.	N/A	N/A
2012/0226117	12/2011	Lamego et al.	N/A	N/A
2012/0283524	12/2011	Kiani et al.	N/A	N/A
2012/0319816	12/2011	Al-Ali	N/A	N/A
2013/0023775	12/2012	Lamego et al.	N/A	N/A
2013/0041591	12/2012	Lamego	N/A	N/A
2013/0060147	12/2012	Welch et al.	N/A	N/A
2013/0090537	12/2012	Schemmann et al.	N/A	N/A
2013/0096405	12/2012	Garfio	N/A	N/A
2013/0096936	12/2012	Sampath et al.	N/A	N/A
2013/0243021	12/2012	Siskavich	N/A	N/A
2013/0253334	12/2012	Al-Ali et al.	N/A	N/A
2013/0296672	12/2012	O'Neil et al.	N/A	N/A
2013/0296713	12/2012	Al-Ali et al.	N/A	N/A
2013/0324808	12/2012	Al-Ali et al.	N/A	N/A
2013/0331660	12/2012	Al-Ali et al.	N/A	N/A
2013/0345921	12/2012	Al-Ali et al.	N/A	N/A
2014/0012100	12/2013	Al-Ali et al.	N/A	N/A
2014/0051953	12/2013	Lamego et al.	N/A	N/A
2014/0081175	12/2013	Telfort	N/A	N/A
2014/0120564	12/2013	Workman et al.	N/A	N/A
2014/0121482	12/2013	Merritt et al.	N/A	N/A
2014/0127137	12/2013	Bellott et al.	N/A	N/A
2014/0135588	12/2013	Al-Ali et al.	N/A	N/A
2014/0163344	12/2013	Al-Ali	N/A	N/A
2014/0163402	12/2013	Lamego et al.	N/A	N/A
2014/0166076	12/2013	Kiani et al.	N/A	N/A
2014/0171763	12/2013	Diab	N/A	N/A
2014/0180038	12/2013	Kiani	N/A	N/A
2014/0180154	12/2013	Sierra et al.	N/A	N/A
2014/0180160	12/2013	Brown et al.	N/A	N/A
2014/0187973	12/2013	Brown et al.	N/A	N/A
2014/0213864	12/2013	Abdul-Hafiz et al.	N/A	N/A
2014/0266790	12/2013	Al-Ali et al.	N/A	N/A
2014/0275808	12/2013	Poeze et al.	N/A	N/A
2014/0275835	12/2013	Lamego et al.	N/A	N/A

2014/0275871	12/2013	Lamego et al.	N/A	N/A
2014/0275872	12/2013	Merritt et al.	N/A	N/A
2014/0276115	12/2013	Dalvi et al.	N/A	N/A
2014/0288400	12/2013	Diab et al.	N/A	N/A
2014/0316217	12/2013	Purdon et al.	N/A	N/A
2014/0316218	12/2013	Purdon et al.	N/A	N/A
2014/0316228	12/2013	Blank et al.	N/A	N/A
2014/0323825	12/2013	Al-Ali et al.	N/A	N/A
2014/0323897	12/2013	Brown et al.	N/A	N/A
2014/0323898	12/2013	Purdon et al.	N/A	N/A
2014/0330092	12/2013	Al-Ali et al.	N/A	N/A
2014/0330098	12/2013	Merritt et al.	N/A	N/A
2014/0357966	12/2013	Al-Ali et al.	N/A	N/A
2015/0005600	12/2014	Blank et al.	N/A	N/A
2015/0011907	12/2014	Purdon et al.	N/A	N/A
2015/0012231	12/2014	Poeze et al.	N/A	N/A
2015/0032029	12/2014	Al-Ali et al.	N/A	N/A
2015/0038859	12/2014	Dalvi et al.	N/A	N/A
2015/0057530	12/2014	Roggeveen et al.	N/A	N/A
2015/0073241	12/2014	Lamego	N/A	N/A
2015/0080754	12/2014	Purdon et al.	N/A	N/A
2015/0087936	12/2014	Al-Ali et al.	N/A	N/A
2015/0094546	12/2014	Al-Ali	N/A	N/A
2015/0097701	12/2014	Muhsin et al.	N/A	N/A
2015/0099950	12/2014	Al-Ali et al.	N/A	N/A
2015/0099955	12/2014	Al-Ali et al.	N/A	N/A
2015/0101844	12/2014	Al-Ali et al.	N/A	N/A
2015/0106121	12/2014	Muhsin et al.	N/A	N/A
2015/0112151	12/2014	Muhsin et al.	N/A	N/A
2015/0116076	12/2014	Al-Ali et al.	N/A	N/A
2015/0165312	12/2014	Kiani	N/A	N/A
2015/0196249	12/2014	Brown et al.	N/A	N/A
2015/0216459	12/2014	Al-Ali et al.	N/A	N/A
2015/0238722	12/2014	Al-Ali	N/A	N/A
2015/0245773	12/2014	Lamego et al.	N/A	N/A
2015/0245794	12/2014	Al-Ali	N/A	N/A
2015/0257689	12/2014	Al-Ali et al.	N/A	N/A
2015/0272514	12/2014	Kiani et al.	N/A	N/A
2015/0351697	12/2014	Weber et al.	N/A	N/A
2015/0359429	12/2014	Al-Ali et al.	N/A	N/A
2015/0366507	12/2014	Blank et al.	N/A	N/A
2016/0029932	12/2015	Al-Ali	N/A	N/A
2016/0058347	12/2015	Reichgott et al.	N/A	N/A
2016/0066824	12/2015	Al-Ali et al.	N/A	N/A
2016/0081552	12/2015	Wojtczuk et al.	N/A	N/A
2016/0095543	12/2015	Telfort et al.	N/A	N/A
2016/0095548	12/2015	Al-Ali et al.	N/A	N/A
2016/0103598	12/2015	Al-Ali et al.	N/A	N/A
2016/0143548	12/2015	Al-Ali	N/A	N/A
2016/0166182	12/2015	Al-Ali et al.	N/A	N/A

2016/0166183	12/2015	Poeze et al.	N/A	N/A
2016/0192869	12/2015	Kiani et al.	N/A	N/A
2016/0196388	12/2015	Lamego	N/A	N/A
2016/0197436	12/2015	Barker et al.	N/A	N/A
2016/0213281	12/2015	Eckerbom et al.	N/A	N/A
2016/0228043	12/2015	O'Neil et al.	N/A	N/A
2016/0233632	12/2015	Scruggs et al.	N/A	N/A
2016/0234944	12/2015	Schmidt et al.	N/A	N/A
2016/0270735	12/2015	Diab et al.	N/A	N/A
2016/0283665	12/2015	Sampath et al.	N/A	N/A
2016/0287090	12/2015	Al-Ali et al.	N/A	N/A
2016/0287786	12/2015	Kiani	N/A	N/A
2016/0296169	12/2015	McHale et al.	N/A	N/A
2016/0310052	12/2015	Al-Ali et al.	N/A	N/A
2016/0314260	12/2015	Kiani	N/A	N/A
2016/0324486	12/2015	Al-Ali et al.	N/A	N/A
2016/0324488	12/2015	Olsen	N/A	N/A
2016/0327984	12/2015	Al-Ali et al.	N/A	N/A
2016/0328528	12/2015	Al-Ali et al.	N/A	N/A
2016/0331332	12/2015	Al-Ali	N/A	N/A
2016/0354015	12/2015	Zhang et al.	N/A	N/A
2016/0367173	12/2015	Dalvi et al.	N/A	N/A
2017/0000394	12/2016	Al-Ali et al.	N/A	N/A
2017/0007134	12/2016	Al-Ali et al.	N/A	N/A
2017/0007198	12/2016	Al-Ali et al.	N/A	N/A
2017/0014056	12/2016	Newberry	N/A	N/A
2017/0014083	12/2016	Diab et al.	N/A	N/A
2017/0014084	12/2016	Al-Ali et al.	N/A	N/A
2017/0024748	12/2016	Haider	N/A	N/A
2017/0027456	12/2016	Kinast et al.	N/A	N/A
2017/0042488	12/2016	Muhsin	N/A	N/A
2017/0055851	12/2016	Al-Ali	N/A	N/A
2017/0055882	12/2016	Al-Ali et al.	N/A	N/A
2017/0055887	12/2016	Al-Ali	N/A	N/A
2017/0055896	12/2016	Al-Ali	N/A	N/A
2017/0079594	12/2016	Telfort et al.	N/A	N/A
2017/0086723	12/2016	Al-Ali et al.	N/A	N/A
2017/0100064	12/2016	Van Dorpe et al.	N/A	N/A
2017/0143281	12/2016	Olsen	N/A	N/A
2017/0147774	12/2016	Kiani	N/A	N/A
2017/0156620	12/2016	Al-Ali et al.	N/A	N/A
2017/0173632	12/2016	Ai-Ali	N/A	N/A
2017/0187146	12/2016	Kiani et al.	N/A	N/A
2017/0188864	12/2016	Drury	N/A	N/A
2017/0188919	12/2016	Al-Ali et al.	N/A	N/A
2017/0196464	12/2016	Jansen et al.	N/A	N/A
2017/0196470	12/2016	Lamego et al.	N/A	N/A
2017/0202490	12/2016	Al-Ali et al.	N/A	N/A
2017/0224220	12/2016	Tunnell et al.	N/A	N/A
2017/0224262	12/2016	Al-Ali	N/A	N/A

2017/0228516	12/2016	Sampath et al.	N/A	N/A
2017/0245790	12/2016	Al-Ali et al.	N/A	N/A
2017/0251974	12/2016	Shreim et al.	N/A	N/A
2017/0251975	12/2016	Shreim et al.	N/A	N/A
2017/0258403	12/2016	Abdul-Hafiz et al.	N/A	N/A
2017/0311891	12/2016	Kiani et al.	N/A	N/A
2017/0325728	12/2016	Al-Ali et al.	N/A	N/A
2017/0332976	12/2016	Al-Ali	N/A	N/A
2017/0340293	12/2016	Al-Ali et al.	N/A	N/A
2017/0360310	12/2016	Kiani	N/A	N/A
2017/0367632	12/2016	Al-Ali et al.	N/A	N/A
2018/0008146	12/2017	Al-Ali et al.	N/A	N/A
2018/0014752	12/2017	Al-Ali et al.	N/A	N/A
2018/0028124	12/2017	Al-Ali et al.	N/A	N/A
2018/0042513	12/2017	Connor	N/A	N/A
2018/0042557	12/2017	Park et al.	N/A	N/A
2018/0055385	12/2017	Al-Ali	N/A	N/A
2018/0055390	12/2017	Kiani et al.	N/A	N/A
2018/0055430	12/2017	Diab et al.	N/A	N/A
2018/0064381	12/2017	Shakespeare et al.	N/A	N/A
2018/0069776	12/2017	Lamego et al.	N/A	N/A
2018/0078155	12/2017	Baek et al.	N/A	N/A
2018/0103874	12/2017	Lee et al.	N/A	N/A
2018/0116575	12/2017	Perea et al.	N/A	N/A
2018/0125368	12/2017	Lamego et al.	N/A	N/A
2018/0125430	12/2017	Al-Ali et al.	N/A	N/A
2018/0130325	12/2017	Kiani et al.	N/A	N/A
2018/0132769	12/2017	Weber et al.	N/A	N/A
2018/0132770	12/2017	Lamego	N/A	N/A
2018/0199871	12/2017	Pauley et al.	N/A	N/A
2018/0242926	12/2017	Muhsin et al.	N/A	N/A
2018/0247353	12/2017	Al-Ali et al.	N/A	N/A
2018/0247712	12/2017	Muhsin et al.	N/A	N/A
2018/0256087	12/2017	Al-Ali et al.	N/A	N/A
2018/0296161	12/2017	Shreim et al.	N/A	N/A
2018/0300919	12/2017	Muhsin et al.	N/A	N/A
2018/0310822	12/2017	Indorf et al.	N/A	N/A
2018/0310823	12/2017	Al-Ali et al.	N/A	N/A
2018/0317826	12/2017	Muhsin et al.	N/A	N/A
2019/0015023	12/2018	Monfre	N/A	N/A
2019/0117070	12/2018	Muhsin et al.	N/A	N/A
2019/0200941	12/2018	Chandran et al.	N/A	N/A
2019/0239787	12/2018	Pauley et al.	N/A	N/A
2019/0320906	12/2018	Olsen	N/A	N/A
2019/0374139	12/2018	Kiani et al.	N/A	N/A
2019/0374173	12/2018	Kiani et al.	N/A	N/A
2019/0374713	12/2018	Kiani et al.	N/A	N/A
2020/0021930	12/2019	Iswanto et al.	N/A	N/A
2020/0060869	12/2019	Telfort et al.	N/A	N/A
2020/0111552	12/2019	Ahmed	N/A	N/A

2020/0113435	12/2019	Muhsin	N/A	N/A
2020/0113488	12/2019	Al-Ali et al.	N/A	N/A
2020/0113496	12/2019	Scruggs et al.	N/A	N/A
2020/0113497	12/2019	Triman et al.	N/A	N/A
2020/0113520	12/2019	Abdul-Hafiz et al.	N/A	N/A
2020/0138288	12/2019	Al-Ali et al.	N/A	N/A
2020/0138368	12/2019	Kiani et al.	N/A	N/A
2020/0196877	12/2019	Vo et al.	N/A	N/A
2020/0253474	12/2019	Muhsin et al.	N/A	N/A
2020/0253544	12/2019	Belur Nagaraj et al.	N/A	N/A
2020/0275841	12/2019	Telfort et al.	N/A	N/A
2020/0288983	12/2019	Telfort et al.	N/A	N/A
2020/0321793	12/2019	Al-Ali et al.	N/A	N/A
2020/0329983	12/2019	Al-Ali et al.	N/A	N/A
2020/0329984	12/2019	Al-Ali et al.	N/A	N/A
2020/0329993	12/2019	Al-Ali et al.	N/A	N/A
2020/0330037	12/2019	Al-Ali et al.	N/A	N/A
2021/0022628	12/2020	Telfort et al.	N/A	N/A
2021/0104173	12/2020	Pauley et al.	N/A	N/A
2021/0113121	12/2020	Diab et al.	N/A	N/A
2021/0117525	12/2020	Kiani et al.	N/A	N/A
2021/0118581	12/2020	Kiani et al.	N/A	N/A
2021/0121582	12/2020	Krishnamani et al.	N/A	N/A
2021/0161465	12/2020	Barker et al.	N/A	N/A
2021/0236729	12/2020	Kiani et al.	N/A	N/A
2021/0256267	12/2020	Ranasinghe et al.	N/A	N/A
2021/0256835	12/2020	Ranasinghe et al.	N/A	N/A
2021/0275101	12/2020	Vo et al.	N/A	N/A
2021/0290060	12/2020	Ahmed	N/A	N/A
2021/0290072	12/2020	Forrest	N/A	N/A
2021/0290080	12/2020	Ahmed	N/A	N/A
2021/0290120	12/2020	Al-Ali	N/A	N/A
2021/0290177	12/2020	Novak, Jr.	N/A	N/A
2021/0290184	12/2020	Ahmed	N/A	N/A
2021/0296008	12/2020	Novak, Jr.	N/A	N/A
2021/0330228	12/2020	Olsen et al.	N/A	N/A
2021/0386382	12/2020	Olsen et al.	N/A	N/A
2021/0402110	12/2020	Pauley et al.	N/A	N/A
2022/0026355	12/2021	Normand et al.	N/A	N/A
2022/0039707	12/2021	Sharma et al.	N/A	N/A
2022/0053892	12/2021	Al-Ali et al.	N/A	N/A
2022/0071562	12/2021	Kiani	N/A	N/A
2022/0096603	12/2021	Kiani et al.	N/A	N/A
2022/0151521	12/2021	Krishnamani et al.	N/A	N/A
2022/0218244	12/2021	Kiani et al.	N/A	N/A
2022/0287574	12/2021	Telfort et al.	N/A	N/A
2022/0296161	12/2021	Al-Ali et al.	N/A	N/A
2022/0361819	12/2021	Al-Ali et al.	N/A	N/A
2022/0379059	12/2021	Yu et al.	N/A	N/A

2022/0392610	12/2021	Kiani et al.	N/A	N/A
2023/0028745	12/2022	Al-Ali	N/A	N/A
2023/0038389	12/2022	Vo	N/A	N/A
2023/0045647	12/2022	Vo	N/A	N/A
2023/0058052	12/2022	Al-Ali	N/A	N/A
2023/0058342	12/2022	Kiani	N/A	N/A
2023/0069789	12/2022	Koo et al.	N/A	N/A
2023/0087671	12/2022	Telfort et al.	N/A	N/A
2023/0110152	12/2022	Forrest et al.	N/A	N/A
2023/0111198	12/2022	Yu et al.	N/A	N/A
2023/0115397	12/2022	Vo et al.	N/A	N/A
2023/0116371	12/2022	Mills et al.	N/A	N/A
2023/0135297	12/2022	Kiani et al.	N/A	N/A
2023/0138098	12/2022	Telfort et al.	N/A	N/A
2023/0145155	12/2022	Krishnamani et al.	N/A	N/A
2023/0147750	12/2022	Barker et al.	N/A	N/A
2023/0210417	12/2022	Al-Ali et al.	N/A	N/A
2023/0222805	12/2022	Muhsin et al.	N/A	N/A
2023/0222887	12/2022	Muhsin et al.	N/A	N/A
2023/0226331	12/2022	Kiani et al.	N/A	N/A
2023/0284916	12/2022	Telfort	N/A	N/A
2023/0284943	12/2022	Scruggs et al.	N/A	N/A
2023/0301562	12/2022	Scruggs et al.	N/A	N/A
2023/0346993	12/2022	Kiani et al.	N/A	N/A
2023/0368221	12/2022	Haider	N/A	N/A
2023/0371893	12/2022	Al-Ali et al.	N/A	N/A
2023/0389837	12/2022	Krishnamani et al.	N/A	N/A
2024/0016418	12/2023	Devadoss et al.	N/A	N/A
2024/0016419	12/2023	Devadoss et al.	N/A	N/A
2024/0047061	12/2023	Al-Ali et al.	N/A	N/A
2024/0049310	12/2023	Al-Ali et al.	N/A	N/A
2024/0049986	12/2023	Al-Ali et al.	N/A	N/A
2024/0081656	12/2023	DeJong et al.	N/A	N/A
2024/0122486	12/2023	Kiani	N/A	N/A
2024/0180456	12/2023	Al-Ali	N/A	N/A
2024/0188872	12/2023	Al-Ali et al.	N/A	N/A
2024/0245855	12/2023	Vo et al.	N/A	N/A
2024/0260894	12/2023	Olsen	N/A	N/A
2024/0267698	12/2023	Telfort et al.	N/A	N/A
2024/0277233	12/2023	Al-Ali	N/A	N/A
2024/0277280	12/2023	Al-Ali	N/A	N/A
2024/0298920	12/2023	Fernkbist et al.	N/A	N/A
2024/0306985	12/2023	Vo et al.	N/A	N/A
2024/0324953	12/2023	Telfort	N/A	N/A
2024/0380246	12/2023	Moran	N/A	N/A
2024/0380247	12/2023	Moran	N/A	N/A
2024/0404549	12/2023	Campbell et al.	N/A	N/A
2025/0000458	12/2024	Abdul-Hafiz et al.	N/A	N/A
2025/0037836	12/2024	Kiani	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3 056 141	12/2015	EP	N/A
WO 2019/014629	12/2018	WO	N/A

OTHER PUBLICATIONS

US 2022/0192529 A1, 06/2022, Al-Ali et al. (withdrawn) cited by applicant
US 2024/0016391 A1, 01/2024, Lapotko et al. (withdrawn) cited by applicant
International Search Report and Written Opinion received in PCT Application No.
PCT/US2018/042148, dated Nov. 19, 2018. cited by applicant

Primary Examiner: Holmes; Rex R

Attorney, Agent or Firm: Knobbe, Martens, Olson & Bear, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 17/324,396, entitled “MEDICAL MONITORING DEVICE FOR HARMONIZING PHYSIOLOGICAL MEASUREMENTS” filed May 19, 2021 and issued as U.S. Pat. No. 11,992,311, which is itself a continuation of U.S. patent application Ser. No. 16/034,743, entitled “MEDICAL MONITORING DEVICE FOR HARMONIZING PHYSIOLOGICAL MEASUREMENTS” filed Jul. 13, 2018 and issued as U.S. Pat. No. 11,026,604, which claims priority benefit to U.S. Provisional Application No. 62/532,273, entitled “MEDICAL MONITORING DEVICE FOR COORDINATING PHYSIOLOGICAL MEASUREMENTS,” filed Jul. 13, 2017, and U.S. Provisional Application No. 62/667,983, entitled “MEDICAL MONITORING DEVICE FOR HARMONIZING PHYSIOLOGICAL MEASUREMENTS,” filed May 7, 2018, each of which is hereby incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) The present disclosure relates to blood glucose determination in biological tissues. Specifically, this disclosure relates to systems, methods, and apparatuses for harmonizing data from a plurality of non-invasive sensors to estimate blood glucose levels.

BACKGROUND

(2) Monitoring of blood glucose (blood sugar) concentration levels has long been critical to the treatment of diabetes in humans. Current blood glucose monitors involve a chemical reaction between blood serum and a test strip, requiring an invasive extraction of blood via a lancet or pinprick. Small handheld monitors have been developed to enable a patient to perform this procedure anywhere, at any time. But the inconvenience of this procedure—specifically the blood extraction and the use and disposition of test strips—has led to a low level of compliance. Such low compliance can lead to serious medical complications. Thus, a non-invasive method for monitoring blood glucose is needed.

SUMMARY

(3) The present disclosure describes example systems, methods, apparatuses, and medical devices for harmonizing data from a plurality of non-invasive sensors. In general, a physiological parameter can be determined by harmonizing data between two or more different types of non-invasive physiological sensors interrogating the same or proximate measurement sites. In some cases, data from one or more first non-invasive sensors can be utilized to identify one or more

variables that are useful in one or more calculations associated with data from one or more second non-invasive sensors. In some cases, data from one or more first non-invasive sensors can be utilized to calibrate one or more second non-invasive sensors. Non-invasive sensors can include, but are not limited to, an optical coherence tomography (OCT) sensor, a bio-impedance sensor, a tissue dielectric constant sensor, a plethysmography sensor, or a Raman spectrometer.

(4) For purposes of summarizing the disclosure, certain aspects, advantages and novel features are discussed herein. It is to be understood that not necessarily all such aspects, advantages or features will be embodied in any particular embodiment of the invention and an artisan would recognize from the disclosure herein a myriad of combinations of such aspects, advantages or features.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The following drawings and the associated descriptions are provided to illustrate embodiments of the present disclosure and do not limit the scope of the claims.

(2) FIG. 1 illustrates an example patient monitoring system that includes a plurality of physiological sensors.

(3) FIG. 2 illustrates a block diagram of an example patient monitoring system.

(4) FIG. 3A illustrates an example 3D OCT image obtained from a volar side of forearm skin.

(5) FIG. 3B illustrates an example one-dimensional distribution of light intensity versus depth graph obtained by averaging scans of the image of FIG. 3A.

(6) FIG. 4 illustrates example micro-invasive elements of a bioimpedance sensor.

(7) FIG. 5 illustrates an example bioimpedance sensor.

(8) FIG. 6 shows a graph illustrating various example light intensity signals acquired at a patient's wrist.

(9) FIG. 7 illustrates a scaled view of the various example light intensity signals of FIG. 6.

(10) FIG. 8 illustrates an approximation of an intensity of the fluorescence portion of the light intensity signals of FIG. 7.

(11) FIG. 9 illustrates an approximation of an intensity of the isolated Raman with tissue absorption signals of FIG. 7.

(12) FIG. 10 illustrates an approximation of an intensity of the isolated Raman with tissue absorption signals of FIG. 7.

(13) FIGS. 11A-11C illustrate optical scattering differences in skin geometries among various age groups.

(14) FIGS. 12A-12B illustrate an example sensor fusion apparatus configured with multiple sensing capabilities that interrogation of the same or a sufficiently proximate tissue site.

(15) FIG. 13 illustrates an example reflectance sensor or probe.

(16) FIG. 14 illustrates an environment that shows a hand of a user interacting with the example reflectance sensor of FIG. 13.

(17) FIG. 15 illustrates an example physiological monitoring system.

(18) FIGS. 16A, 16B, and 16C illustrate an example physiological monitoring apparatus.

(19) FIG. 17 illustrates a flow diagram illustrative of an example routine for harmonizing data from a plurality of non-invasive sensors.

(20) FIG. 18 illustrates a flow diagram illustrative of an example routine for harmonizing data from a plurality of non-invasive sensors.

(21) While the foregoing "Brief Description of the Drawings" references generally various embodiments of the disclosure, an artisan will recognize from the disclosure herein that such embodiments are not mutually exclusive. Rather, the artisan would recognize a myriad of combinations of some or all of such embodiments.

DETAILED DESCRIPTION

(22) Overview

(23) Many non-invasive techniques for determining blood glucose have significant shortcomings, such as low accuracy (for example, less accuracy than invasive home monitors) and insufficient specificity of glucose concentration measurement. Accordingly, there is a need for an improved method to non-invasively monitor glucose. Systems and methods disclosed herein address various challenges related to non-invasively determining a patient's blood glucose level by harmonizing data from multiple non-invasive sensors. Each of the non-invasive sensors can interrogate the same or a similar tissue site of a patient, and variables identified using one or more sensors can be utilized to improve data from one or more other sensors. Using these data harmonization techniques, a glucose concentration measurement can be obtained.

(24) In many instances, a single non-invasive sensor may lack the functionality to measure each of the parameters required for an accurate determination of an analyte concentration. As a result, many physiological monitoring techniques include estimations, such as those based on common assumptions, to compensate for the lack of known data. However, due to the sensitivity of analyte measurements, these estimations can result in inaccurate or unreliable determinations.

(25) For example, Beer's Law (also known as the Beer-Lambert Law) relates the attenuation of light to properties of a material. In particular, Beer's law states that absorbance of a material is proportional to the concentrations of the attenuating species in the material sample. The relationship between these parameters is expressed in Equation 1 below:

(26) $A = \epsilon * b * c$ (Equation 1)

(27) where A is the absorbance of the material at a given wavelength of light, ϵ is the molar absorptivity or extinction coefficient ($L \cdot mol^{-1} \cdot cm^{-1}$), unique to each molecule and varying with wavelength, b is the length of the light path through the material (cm), and c is the concentration of an analyte of interest ($mol \cdot L^{-1}$).

(28) In many cases, the length of the light path through the material (sometimes referred to as the path length) is estimated. For example, a generic finger may be associated with a first estimated path length value, while a generic nose may be associated with a second path length value. However, every person has a unique tissue geometry, which can include, but is not limited to, unique skin structure or skin thickness. Furthermore, because tissue is not uniform throughout a person's body, even tissue sites that are close in proximity, such as two different measurements sites on a patient's finger, can have a different tissue geometry. As noted above, a specific tissue geometry of a particular tissue site can affect the path length value. Accordingly, a non-invasive physiological sensor can be configured to obtain skin geometry data, which can be utilized to calculate a path length associated with a tissue site. In addition or alternatively, the skin geometry data can be utilized to calibrate one or more sensors (for example, select a focal depth of Raman spectrometer), which can result in more accurate analytes measurements, such as blood glucose concentration measurements.

(29) An optical coherence tomography, or OCT, sensor can be utilized to obtain tissue geometry information. OCT is an optical imaging technique using light waves that produce high-resolution imagery of biological tissue. OCT creates its images by interferometrically scanning in depth a linear succession of spots, and measuring backscattered light at different depths in each successive spot. The OCT data can be processed to present an image of the linear cross section. OCT data can be processed to determine tissue geometry information, such as skin geometry. For example, the OCT data can provide data regarding a thickness of one or more skin layers, such as the epidermis, the dermoepidermal junction, or the dermis.

(30) In addition or alternatively, OCT data can be utilized to determine whether successive OCT measurements have occurred in the same or a different location. For example, one reason data harmonization between sensors is available relates to the specific optical profile of a particular

tissue site. That is, a particular tissue site retains its specific optical profile, and a different measurement location may have a different optical profile. Thus, in many cases, to maintain data harmonization capabilities, each of the sensors should interrogate the same or a substantially proximate tissue site. One problem associated with interrogating the same or a substantially proximate tissue site relates to the subsequent placement of a sensor after it has been removed from the patient. To solve these and other problems, tissue geometry information associated with OCT data can be utilized to determine whether a later one of successive OCT measurements is taken at the same tissue site as a previous one of the successive OCT measurements.

(31) A bio-impedance or tissue dielectric constant sensor can be utilized to obtain tissue geometry information. For example, bio-impedance or tissue dielectric constant data can provide information relating to one or more skin layers, a hydration of one or more skin layers, or a cellular structure of the tissue.

(32) Raman spectroscopy has exhibited promise with respect to blood glucose detection, for example, due to its capability to gain information about the molecular constitution non-invasively. For example, features such as peaks of the Raman spectra are considered the Raman “fingerprints” of analytes such as glucose. Accordingly, using an isolated or semi-isolated Raman signal, the system can identify physiological data, such as information regarding a patient's blood glucose level.

(33) For various reasons, it has been challenging to isolate a pure Raman signal from a signal obtained from a Raman spectrometer. For example, emission of fluorescence in tissue often overwhelms any signal collected from the Raman spectrometer, thereby hiding Raman features. In addition, attenuation of the signal due to absorption can further affect prediction of analytes using the collected signal. Furthermore, varying tissue geometries at tissue sites increases a difficulty in selecting a focal depth of the Raman spectrometer that will optimize a resolution of the Raman signal.

(34) Systems and methods disclosed herein address one or more of these or other challenges by utilizing data associated with one or more sensors to calibrate or improve an accuracy of one or more other sensors. For example, a value for path length can be obtained from skin geometry data, which can improve a pulse oximetry sensor such as a near infrared (NIR), reflectance, or transmittance sensor. As another example, the present disclosure addresses various challenges related to leveraging the Raman scattering signatures for prediction of glucose by harmonizing data from a plurality of non-invasive physiological sensors. For instance, a focal depth of the Raman spectrometer can be selected based on tissue geometry data, which can improve the Raman spectrometer, and possibly increase an accuracy of a blood glucose measurement. Similarly, using data from one or more sensor, the Raman signal can be isolated by reducing or removing an effect of Fluorescence on a collected signal, or removing an effect of attenuation of the signal due to absorption.

(35) System Overview

(36) FIG. 1 illustrates an example patient monitoring system **100** that includes a patient monitor **102**, a first sensor **104A**, and a second sensor **104B**. In addition, the patient monitoring system **100** can include one or more other sensors **104N**. Sensors **104A**, **104B**, and **104N** can interrogate tissue sites **106A**, **106B**, and **106N**, respectively, of a patient. In some cases, tissue sites **106A**, **106B**, and **106N** can be the same or substantially proximate tissue sites, while in other cases one or more of the tissue sites **106A**, **106B**, or **106N** can be different. Sensor data from the sensors **104A**, **104B**, or **104N** can be utilized to determine one or more physiological parameters or patient vitals. For example, the patient monitor **102** can receive a signal from the one or more of the sensors **104A**, **104B**, or **104N** and can determine, based on the received signal(s), one or more physiological parameters or one or more measurements that can be used to determine a physiological parameter.

(37) The sensors **104A**, **104B**, and **104N** can each be the same type of sensors, or one or more of the sensors **104A**, **104B**, and **104N** can be different from each other. For example, the sensors

104A, **104B**, and **104N** can include, but are not limited to, any combination of an optical coherence tomography (OCT) device, a spectrometer (for example, a Raman spectrometer), a plethysmograph sensor such as a pulse oximetry device (for example, a near infrared (NIR), reflectance and/or transmittance device), a pressure sensor, an electrocardiogram sensor, a bioimpedance sensor, or acoustic sensor, among other sensors.

(38) Two or more of the sensors **104A**, **104B**, or **104N** can be configured to interrogate the same tissue site. For example, two or more of the sensor sensors **104A**, **104B**, or **104N** can be positioned proximate each other such that they can interrogate the same tissue, such as a finger, a thumb, a thenar space, a hand, a wrist, a forearm, a nose, a limb, a head, an ear, a neck, an upper body, or a lower body. In addition or alternatively, two or more of the sensors **104A**, **104B**, or **104N** can be configured to interrogate different tissue sites.

(39) In some cases, one or more of the sensors **104A**, **104B**, or **104N** can be integrated into an apparatus, such as an apparatus that is wearable by a user. For example, one or more of the sensors **104A**, **104B**, or **104N** can be integrated into a glove that when worn by a user allows the sensor(s) to interrogate one or more tissue sites. Similarly, one or more of the sensors **104A**, **104B**, or **104N** can be incorporated in or attached to various other apparatuses, including, but not limited to, a sock, a shirt, a sleeve, a cuff, a bracelet, a glove, or the like.

(40) In some cases, data from a single sensor **104A**, **104B**, or **104N** does not provide enough reliable information to determine certain physiological parameters. For example, a number of factors can affect an accuracy of sensor data including, but not limited to, patient movement, sensor placement, interference, the type of sensor being used, the expansion and contraction of the patient's vascular system, assumptions made during calculations, skin temperature, pressure, or the like. In addition or alternatively, the determination of some physiological parameters (for example, glucose concentration) may require more information than a single sensor can provide.

(41) To solve this and other problems, the patient monitor **102** (or one or more of the sensors) can harmonize or compare data from two or more sensors, which can allow for a determination of more accurate or reliable data, or can allow for a determination of one or more additional physiological parameters, such as blood glucose concentration.

(42) As one example, the patient monitor **102** receives a first signal from a first sensor **104A**, the first signal corresponding to an interrogation of the first tissue site **106A** by the first sensor **104A**. Further, the patient monitor **102** receives a second signal from a second sensor **104B**, the second signal corresponding to an interrogation of the second tissue site **106B** by the second sensor **104B**. Based on the first signal, the patient monitor **102** can make adjustments to modify the second sensor or the second measurement to improve the accuracy or reliability of the second sensor or the second measurement. For instance, adjustments can include, but are not limited to, adjusting an intensity, power, position, or timing of the second sensor **104b** or adjusting values corresponding to the measurement of the second physiological parameter. For example, the patient monitor **102** can modify the second measurement or calculations for a physiological parameter (for example, introduce an offset, adjust assumed or estimated values, filter a signal, etc.) to account for information from the first sensor. In addition or alternatively, the patient monitor can adjust a confidence value associated with the first, second, or another measurement.

(43) As described above, based at least in part on the first and second signals, the patient monitor **102** can determine a physiological parameter. The physiological parameter can be a value which may not be independently determinable from data from either of the first sensor or the second sensor alone. For example, data from the first sensor can be utilized to determine a path length, data from the second sensor can be utilized to determine an absorbance, and the physiological parameter can include a concentration of an analyte, such as glucose. As another example, data from the first sensor can be utilized to determine a path length or absorbance, the second sensor can correspond to a Raman spectrometer, and the physiological parameter can include a concentration of an analyte, such as glucose.

(44) The patient monitor **102** can include a digital signal processor (DSP) that receives the signals generated by the one or more sensors **104A**, **104B**, or **104N** (for example, through a front-end unit) and determines parameters, for example, those indicative of the physiological condition of the patient, using the received signals. The patient monitor **102** can, for example, determine physiological parameters corresponding to the patient, such as an amount of light absorbed, transmitted through, or reflected at a tissue site, path length (for example, distance that light travels through the material), concentration of an analyte, bioimpedance, tissue dielectric constant, pulse rate (PR), pulse pressure variation (PPV), pleth variability index (PVI®), stroke volume (SV), stroke volume variation (SVV), peripheral capillary oxygen saturation (SpO₂), mean arterial pressure (MAP), central venous pressure (CVP), pulse pressure (PP), perfusion index (PI), total hemoglobin (SpHb®), carboxyhemoglobin (SpCO®), methemoglobin (SpMet®), oxygen content (SpOC®), or acoustic respiration rate (RRa®), among other parameters. In some aspects, the patient monitor **102** can derive or use one or more relationships (for instance, a set of linear equations) from two or more of the determined parameters. The patient monitor **102** can utilize the one or more relationships to determine the patient's glucose levels, systemic vascular resistance (SVR), CO, or arterial blood pressure (BP), among other parameters.

(45) The patient monitor **102** can further compare or analyze one or more of the determined parameters (for instance, at least two of the determined parameters or one determined parameter and a previous or model parameter) to adjust how a parameter is measured or calculated to make the measured parameter more accurate or reliable, to adjust a sensor to make the measured parameter more accurate or reliable, to calculate, derive or determine an accuracy or a confidence value of a measured parameter, to isolate a parameter, or to determine another parameter based on the one or more parameters. The sensors, in addition to or alternatively than the patient monitor, can coordinate with each other to coordinate data or adjust calculations to enhance an accuracy or reliability of measurements. In addition or alternatively, the patient monitor **102** can use the data to increase an accuracy of one or more calculations, calculate a previously unknown or estimated physiological parameter, calibrate data, or compensate for various circumstances that might otherwise result in inaccurate or unreliable data.

(46) Additional Implementations

(47) The patient monitor **102** can be connected to one or more (for instance, three, four, five, or six) sensors, such as the sensors **104A**, **104B**, or **104N**, that are detecting from a patient and use the signals received from the sensors to determine one or more physiological parameters including, but not limited to, glucose, SpO₂, PPR, PVI® (for instance, via a palm, thumb or finger plethysmography sensor), SV, MAP, CVP, PP, or PI (for instance, via a palm, thumb or finger plethysmography sensor), among other parameters such as those described herein.

(48) Moreover, the patient monitor **102** can utilize any of the techniques described herein to determine whether any measurement described herein (using any of the sensors described herein) is valid. The patient monitor **102** can be configured to show (for example, on a display) information about a valid or invalid measurement, activate an indicator light (such as an LED), trigger an alarm, adjust one or more sensors or parameters (for instance, based on a received sensor signal), or display any data.

(49) The patient monitor **102** can wirelessly or using wires receive, via an input of the patient monitor **102**, a signal from one of the sensors **104A**, **104B**, or **104N**. The received signal may take various forms, such as a voltage, a current, or charge. An operational amplifier (op-amp) of the patient monitor **102** can increase the amplitude, as well as transform the signal, such as from a current to a voltage. An anti-aliasing filter (AAF) of the patient monitor **102** can then process of the output signal from the op-amp to restrict a bandwidth of the output signal from the op-amp to approximately or completely satisfy the sampling theorem over a band of interest. An analog-to-digital convertor (ADC) of the patient monitor **102** can convert the output signal from the AAF from analog to digital. The output signal from the ADC can then be sampled by a first processor of

the patient monitor **102** at a relatively high speed. The result of the sampling can next be downsampled by a second processor of the patient monitor **102**, which may be the same or different from the first processor, before waveform analysis may be performed by a DSP.

(50) FIG. 2 illustrates a block diagram of an example patient monitoring system **200**, which can be an embodiment of the patient monitoring system **100**. The patient monitoring system **200** can include a patient monitor **202**, a first non-invasive physiological sensor **204A**, a second non-invasive physiological sensor **204B**, or a third non-invasive physiological sensor **204C**. Furthermore, it should be noted that fewer, additional, or different sensors may be included in patient monitoring system **200**.

(51) The sensors **204A**, **204B**, or **204C** can respectively detect from tissue sites **206A**, **206B**, and **206C** of a patient. Each of the sensor can measure from the same or a similar tissue site. For example, sensor **204A** can take a measurement and sensor **204B** can take a subsequent measurement on the same tissue or at the same location. This may allow the system to more easily harmonize the data from the sensors or use data from one sensor to improve data or calculation based on another sensor. The tissue sites **206A**, **206B**, and **206C** can be different. As a non-limiting example, tissue site **206A** can include a thenar space of a patient's hand, and tissue sites **206B**, **206C** include a thumb of the patient, such as a base of the thumb. It should be noted, however, that fewer, more or different sensors can be include in system **200**.

(52) The DSP **212A** can communicate via drivers **216A** with the plethysmography sensor **204A** and receive via a front-end **214A** one or more light intensity signals indicative of one or more physiological parameters of the patient or one or more measurements that can be used to determine one or more physiological parameters. For example, a signal can be indicative of an intensity of light reflected, refracted, scattered, absorbed, or transmitted at a tissue site. The drivers **216A** can convert digital control signals into analog drive signals capable of driving emitters **209A** to illuminate the tissue site **206A**. For example, the light emitted by emitters **209A** can have an infrared (IR), near infrared (NIR), red, ultra-violet (UV), visible, or other wavelength. The detector(s) **208A** can, in turn, generate one or more composite analog light intensity signals responsive to light detected by the detector(s) **208A** after attenuation, reflection, refraction, scattering, absorption, etc. at the tissue site **206A**. The emitter(s) **209A** or detector(s) **208A** include a fiber-optic component for illumination and collection, respectively. For example, the emitter(s) **209A** can be positioned on a tissue site **206A** (for example, on top, on the bottom, on the side, etc.) and the detector(s) **208A** can be positioned on an opposite portion of the tissue site **206A**.

(53) The front-end **214A** can convert the one or more composite analog light intensity signals from the detector(s) **208A** into digital data and input the digital data into the DSP **212A**. The digital data from the front-end **216A** can correspond to at least one of a plurality of physiological parameters as described herein. For example, the digital data from the front-end **216A** can be representative of a change in the absorption of particular wavelengths of light as a function of the changes in the tissue site **206A** resulting from pulsing blood.

(54) The DSP **212A** can include one or more data or signal processors configured to execute one or more programs for determining physiological parameters from input data. The DSP **212A** can perform operations that include calculating or outputting one or more physiological measures, such as absorbance, path length, PVI® and other parameters described herein. The operations performed by the DSP **212A** can be implemented in software, firmware or other form of code or instructions, or logic or other hardware, or a combination of the above.

(55) The instrument manager **210** can communicate with one or more input or output devices **220**. The one or more input or output devices **220** can include a user interface **222**, controls **224**, a transceiver **226**, and a memory device **228**.

(56) The user interface **222** can include a numerical or graphical display that provides readouts of measures or parameters, trends and bar graphs of measures or parameters, visual indications of measures or parameters, visual indicators like LEDs of various colors that signify measurement

magnitude, or device management interfaces, which can be generated by LEDs, LCDs, or CRTs, for example. The user interface **222** can include an audible output device that provides readouts or audible indications of measures or parameters. The user interface **222** can include one or more input devices like a keypad, touch screen, pointing device, voice recognition device, and computer that can be used to supply control or configuration data, such as initialization settings, from the user interface **222** to the instrument manager **210**. In some implementations, the user interface **222** can be an interface for devices as well as users.

(57) The controls **224** can be outputs to medical equipment, such as drug administration devices, ventilators, or fluid IVs, so as to control the amount of administered drugs, ventilator settings, or the amount of infused fluids. The patient monitor **202** can use the controls **224** to automatically treat the patient (for instance, provide fluid to the patient, provide medication to the patient, turn on a fan to cool the patient, or adjust a temperature of a room to heat or cool the patient) in response to determining that the patient may benefit from treatment.

(58) The transceiver **226** via an antenna can transmit information about operation of the patient monitor **202** to an electronic device or receive control or configuration data for operating the patient monitor **202**. The transceiver can, for example, communicate via a computer network or intermediary device or directly with the electronic device using electromagnetic radiation.

(59) The memory device **228** can be used to store information about operation of the patient monitor **202**. This information can, for example, include readouts of measures or parameters, trends and bar graphs of measures or parameters, visual indications or indicators.

(60) The DSP **212B** can receive via a front-end **214B** one or more light intensity signals indicative of one or more physiological parameters of the patient. The drivers **216B** can convert digital control signals into analog drive signals capable of driving emitters/detectors **220** to illuminate the tissue site **206B**. For example, the light emitted by emitters/detectors **220** can be infrared (IR), near infrared (NIR), red, ultra-violet (UV), visible, or other wavelength. The emitters/detectors **220** can, in turn, generate one or more composite analog light intensity signals responsive to light detected by the emitters/detectors **220** light is reflected, refracted, scattered, absorbed, or attenuated at a tissue site **206B**. The emitters/detectors **220** include a fiber-optic bundle that has illumination and detection fibers. In addition, for example, as described with respect to FIG. **1**, the emitters/detectors **220** can be separate.

(61) The front-end **214B** can convert the one or more composite analog light intensity signals from the emitters/detectors **220** into digital data and input the digital data into the DSP **212B**. The digital data from the front-end **214B** can correspond to at least one of a plurality of physiological parameters, as described herein. The digital data from the front-end **214B** can be representative of a change in the absorption/reflection of particular wavelengths of light as a function of the changes in the tissue site **206B** resulting from pulsing blood.

(62) The DSP **212B** can include one or more data or signal processors configured to execute one or more programs for determining physiological parameters from input data. The operations performed by the DSP **212B** can be implemented in software, firmware or other form of code or instructions, or logic or other hardware, or a combination of the above.

(63) Sensor **204C** includes a detector **208C**, a light source **209C**, a beam splitter **224C**, and a reflector **222C**. The light source **209C** can emit light having an approximately equal wavelength, a spectrum of wavelengths, or a few different wavelengths, for example, two. For example, the wavelengths can be selected based on the absorption spectrum.

(64) As illustrated, light beams from the light source **209C** are split using the beam splitter **224C** into reference arm light beams **230** and sample arm light beams **228**. After the light beams **234** are split, the reference arm light beams **230** travel down the reference arm to interact with the reflector **222C**, and the sample arm light beams **228** travel down the sample arm to interact with the tissue **206C**, for example, from the base of a patient's thumb.

(65) The tissue site **206C** can absorb, reflect, scatter, or refract the sample arm light beams **228**.

Some of the sample arm light beams **228** are reflected back to the beam splitter **224C**. The beam splitter **224C** can direct at least some of the reflected sample arm light beams **228** to the detector **208C**.

(66) The light beams traveling down the reference arm interact with a reflector **222C** and are reflected back to the beam splitter **224C**. Similar to the reflected sample arm light beams **228**, the reflected reference arm light beams **230** are also directed to the detector **208C** by the beam splitter **224C**. Reflected signals from the sample arm and reference arm and are presented to photodetector **208C** for measurement.

(67) The tissue volume with which the light interacts (referred to as the interaction volume) can be determined by the spot size of the imaging optics (surface area) and the coherence length of the light (depth). Thus, the reference arm can determine the depth within the interaction volume from which scattered light is measured. The patient monitor **200** uses the detected signals obtained from the interference of the reflected sample arm light beams **228** and the reflected reference arm light beams **230** to calculate tissue geometry data, such as a skin geometry of one or more skin layers.

(68) Although not illustrated in FIG. 2, imaging optics can also be used to focus the sample arm light beams **228** prior to interacting with the tissue site **206C**. Furthermore, the end of the sample arm and imaging optics can be placed in close proximity to the tissue site **206C**. The reference arm and reflector **222** are configured such that appropriate wavelength and polarization selected such that the appropriate depth of the tissue is measured.

(69) The DSP **212C** can receive via a front-end **214C** one or more signals indicative of one or more physiological parameters of the patient, such as path length. The drivers **216C** can convert digital control signals into analog drive signals capable of driving emitters **209C** to illuminate the tissue site **206C**. The detectors **208C** can, in turn, generate one or more composite analog signals responsive to light detected by the detectors **208C**.

(70) The front-end **214C** can convert the one or more composite analog signals from the detectors **208C** into digital data and input the digital data into the DSP **212C**. The digital data from the front-end **216C** can correspond to at least one of a plurality of physiological parameters, as described herein. The DSP **212C** can include one or more data or signal processors configured to execute one or more programs for determining physiological parameters from input data. The operations performed by the DSP **212C** can be implemented in software, firmware or other form of code or instructions, or logic or other hardware, or a combination of the above.

(71) One or more of the components relating to signal acquisition or processing (for example, front end **214A**, **214B**, **214C**, drivers **216A**, **216B**, **216C**, DSP **212A**, **212B**, **212C**, etc.) can be incorporated into one or more connecting cables, the sensors themselves, or are otherwise closer to the sensor sites. As such, the patient monitor **202** can include primarily the input or output devices **220** and the instrument manager **210**, (if appropriate). In addition, some of the components are illustrated as separate units but can be combined. For instance, front end **214A**, **214B**, **214C** can be combined into one or more front ends, drivers **216A**, **216B**, **216C**, can be combined into one or more drives, DSP **212A**, **212B**, **212C** can be combined into one or more DSPs, etc. By reducing the number of components included in the patient monitor **102**, **202**, the monitor can be smaller in size or more portable, which can be more convenient for home or “spot check” use.

(72) Although not illustrated in FIG. 1 or 2 patient monitors **102**, **202**, or cables connecting the patient monitors to the sensors can further include one or more outputs that supply the signal(s) from one or more of the sensors to one or more other electronic devices for further processing. As one example, the signal(s) from one or more of the sensors can be output in parallel by one or more of the sensors or the cables that couple the one or more sensors to the patient monitor **102**, **202**. In another example, the patient monitors **102**, **202** can include one or more outputs for outputting copy(ies) of the signal(s) from one or more of the sensors. The copy(ies) of the signal(s) can also be adjusted relative to the original(s) with filtering, scaling, or other changing prior to being provided to the one or more other electric devices.

(73) Optical Coherence Tomography

(74) Optical coherence tomography, or OCT, is an optical imaging technique using light waves that produces high-resolution imagery of biological tissue. OCT creates its images by focusing a beam of light into a medium and interferometrically scanning the depth of a linear succession of spots and measuring the absorption and/or the scattering of the light at different depths in each successive spot. In some cases, the data can be processed to present an image of the linear cross section of the medium scanned.

(75) A light source can output a beam of light having a broad spectrum of wavelengths. The beam of light can be collimated and pass a beam splitter such that a portion of the beam of light is directed towards the tissue and a portion of the beam of light is directed toward a reference arm. The light can be either polarized or non-polarized. A polarizer located on one edge of the beam splitter can polarize the light linearly, elliptically, or circularly, as desired. The path length of the reference arm can be changed based on the desired measurement depth into the tissue. The wavelength can be centered at, for example, 1310 nm with a 50 nm bandwidth. In other cases, the wavelength can be 1060 nm with a 70 nm bandwidth. The light source can be selected to have a center wavelength anywhere between 400 nm and 1700 nm with a bandwidth of up to 150 nm. It is understood that different light sources with different bandwidths can be chosen to optimize penetration depth into the tissue and optimize the depth resolution of sensitivity to skin structures. The reflected light from the tissue can be collected using a converging lens and be directed back through the beam splitter to a photodetector where it is recombined with a portion of the reference arm beam to form an interference pattern. A processor can use the signals from the photodetector to render an image of the tissue.

(76) OCT can provide a non-invasive method for identifying one or more characteristics of a tissue's structure. For example, OCT data (which can be referred to as tissue geometry data) can include an indication of a boundary between the main skin layers, such as the epidermis (outermost layer of the skin), the dermis (layer beneath the epidermis), or the hypodermis (layer directly below the dermis and serves to connect the skin to the underlying fibrous tissue of the bones or muscles). The epidermis is further divided into five, separate layers (Stratum Corneum, Stratum Lucidum, Stratum Granulosum, Stratum Spinosum, and Stratum Basale) and the dermis is divided into two, separate layers (the papillary dermis and the reticular dermis). In some cases, OCT data can provide an indication of a boundary between any of these layers. In addition or alternatively, OCT data can include an indication of a thickness of any of the epidermis, dermis, or hypodermis, or their individual layers.

(77) For example, FIG. 3A illustrates an example 3D OCT image obtained from a volar side of forearm skin, and FIG. 3B illustrates an example one-dimensional distribution of light intensity vs. depth obtained by averaging Amplitude scans (A-scans) in the reconstructed OCT 3D image of FIG. 3A. The slope of the line of FIG. 3B is indicative of index of refraction of tissue. A difference in the index of refraction, or a difference in slope, can indicate a new skin or tissue layer because each layer may have a different index of refraction. As illustrated, the first peak **302** corresponds to the skin surface **302**, and the second peak **308** corresponds to the dermoepidermal junction, which is the area of tissue that joins the epidermis **310** and the dermis layers (for example, the papillary dermis **304**) of the skin. Accordingly, using OCT data, the system **200** can determine a thickness of one or more of various skin layers such as, but not limited to, the epidermis **310**, the dermoepidermal junction, the papillary dermis **304**, the reticular dermis **306**, or the like.

(78) In some cases, OCT data can provide an indication that an OCT sensor is interrogating an unfavorable tissue site. An unfavorable tissue site can include any tissue site that might provide distorted or inaccurate OCT data (relative to desired OCT data), such as tissue sites that include at least a portion of a hair follicle, a pore, a bone, a finger- or toe-nail, a pimple, a mole, a scar, a blister, a callous, debris, other skin imperfection, or the like.

(79) A particular tissue site can retain its specific optical profile over time, and that optical profile

can be different from another tissue site. Accordingly, to maintain data harmonization capabilities, it can be advantageous for sensors to interrogate the same or a substantially proximate tissue site. One problem associated with interrogating the same or a substantially proximate tissue site relates to the subsequent placement of a sensor after it has been removed from the patient or when it is shifted in some way from its original positioning. For example, a subsequent OCT measurement or set of measurements can occur minutes, hours, days, weeks, or some other period of time after the first measurement, and it can be unreasonable to require a patient to wear or interact with the OCT sensor for the duration of that period of time. Nonetheless, even though the OCT sensor has been separated from the patient or shifted from its original position, it can be advantageous for the subsequent OCT measurement(s) to occur at the same location as the first measurement. For example, as described herein, a first tissue site may have a different tissue structure, density, depth, hydration, analyte concentration, or the like than a second, different tissue site. Thus, if the OCT sensor is placed at the same location for each measurement, then previous calculations, determinations, or the like can be utilized, which can simplify any calibrations or corrections to sensor data, among other things.

(80) To solve these and other problems, tissue geometry information associated with OCT data can be utilized to determine whether a subsequent placement of the OCT sensor allows the OCT sensor to interrogate the tissue site corresponding to the tissue site of the first OCT measurement(s). For example, a processor can compare the first tissue geometry data associated with the first OCT measurement(s) with the subsequent tissue geometry data associated with the subsequent OCT measurement(s). If the subsequent tissue geometry data does not correspond to the first tissue geometry data, then the processor can cause one or more actions to occur. For example, the processor can cause an output to indicate that the subsequent tissue geometry data does not correspond to the first tissue geometry data. In other words, the processor can cause an output to indicate that the subsequent placement of the OCT sensor is incorrect, or is different from the first OCT sensor placement, or the processor can cause an output to indicate a probe-off condition. In addition or alternatively, the processor can cause the OCT sensor to be re-positioned. For example, based on the comparison, the processor can suggest a new placement of the OCT sensor, which may more closely correspond to the first placement of the OCT sensor. In addition or alternatively, the processor can control a motorized component to re-position the OCT sensor such that it more closely corresponds to the first placement of the OCT sensor. Still, in some implementations, the processor can calibrate other sensors based on the subsequent tissue geometry data, rather than the first tissue geometry data.

(81) Alternatively, if the subsequent tissue geometry data does correspond to the first tissue geometry data, then the processor can cause one or more other actions to occur. For example, the processor can cause an output to indicate that the subsequent tissue geometry data does correspond to the first tissue geometry data. In other words, the processor can cause an output to indicate that the subsequent placement of the OCT sensor is correct, as compared to the first placement of the OCT sensor. In addition or alternatively, the processor can calibrate other sensors based on the first tissue geometry data or the subsequent tissue geometry data.

(82) Bioelectrical Impedance (Bioimpedance)

(83) Impedance can be characterized as a physical variable describing the resistance characteristics acting on an electric current. Bioelectrical impedance is based on the principle that tissues or fluids of a patient have different impedances, that is, opposition to the flow of the electric current, which in turn may be dependent on variables such as water and electrolyte content, to name a few. Using a bioelectrical impedance, analysis can be performed to examine electrical, capacitive, or resistive characteristics of tissue to provide information on a noninvasive basis.

(84) Mathematically, bioelectrical impedance can be represented as a complex number including a real component (resistance) and an imaginary dimension (reactance). For example, the bioelectrical impedance can be calculated using the following equation below:

(85) $Z = R + jX = \text{.Math. } Z \text{ .Math. } e^{j\theta}$ (Equation2)

where R is resistance, X is reactance, |Z| is amplitude, and θ is phase.

(86) A number of physiological characteristics or parameters can be calculated or estimated using determined bioelectrical impedance characteristics, such as water content, body cell mass (BCM), extra cellular mass (ECM), extracellular fluid (ECF), extracellular water (ECW), fat-free mass (FFM), fat mass (FM), total body water (TBW), electrolyte composition, cell membrane mass, cell membrane function and the like.

(87) Biological tissues can have complex electrical impedance which is dependent, for instance, on the frequency of the electrical applied field or tissue cellular structure. Therefore, the electrical impedance of tissue is a function of its structure and it can be used to differentiate or determine characteristics of one or more layers to tissue.

(88) The system can include a bioimpedance sensor configured to apply an electrical signal to the tissue, which can include one or more of various voltages, currents, frequencies (for example, 1 kHz to 2.5 GHz), or fields. In some cases, the path length of the signal can vary based on the applied electrical signal. For example, low frequency signals may primarily reflect the extracellular environment, while high frequency signals may reflect both the intra- and extracellular environment. In addition, the bioimpedance sensor can be configured to measure characteristics of the applied electrical signal as it passes (or after it has passed) through tissue. For example, the bioimpedance sensor can measure a voltage, current, frequency, magnetic field, etc., which can be indicative of a voltage difference across tissue or a biological impedance of a tissue, to name a few.

(89) One or more properties of skin may disturb or disrupt bioimpedance measurements. For example, the stratum corneum can limit bioimpedance measurements. Accordingly, as illustrated in FIG. 4, the bioimpedance sensor can include a micro-invasive element **402** that is configured to penetrate the stratum corneum layer. For example, the bioimpedance sensor can include spikes or other elements that penetrate approximately 10-20 μm deep.

(90) FIG. 5 illustrates an example bioimpedance sensor **502**. The sensor **502** can include multiple channels of spiked regions configured to penetrate the skin. As shown, spacing between the channels can allow for shallow and deep penetration, such that the bioimpedance sensor **502** can measure impedance at various depths, such as Depths **1a**, **2a**, **2b**, **3a**, or **4a**.

(91) Using information from the bioelectric sensor(s) **502**, the system **200** can determine information about the tissue geometry. For example, based on bioelectric sensor data, the system can determine a cellular structure of the tissue, which may affect various physiological parameters, such as path length or absorption. In addition, based on bioelectric sensor data, the system can determine information related to hydration of the skin or tissue. For example, water content can be directly related to skin thickness. As described herein, in some cases, the system can select a focal depth of the Raman spectrometer based at least in part on tissue geometry data.

(92) Tissue Dielectric Constant

(93) In addition or alternatively to bioimpedance or OCT, the system can utilize one or more tissue dielectric constant sensors to determine various tissue geometries or tissue information, including, but not limited to a dielectric constant of tissue. For example, the system **200** can include a plurality of probes for different measuring depths, such as 0.5 mm, 1.5 mm, 2.5 mm, and 5 mm effective depths, and the system can determine a dielectric value at each of the different depths. In addition or alternatively, the system **200** can include one or more probes that are each configured to measure at different depths, such as 0.5 mm, 1.5 mm, 2.5 mm, and 5 mm effective depths, and the system can determine a dielectric value at each of the different depths. The dielectric value can correlate with water content, which can be tied to tissue structure.

(94) Accordingly, the tissue dielectric constant can provide information which can be combined with other sensor information (for example, OCT, bioimpedance, reflectance or transmission measurements, Raman measurements) to determine more accurate physiological measurements,

such as blood glucose levels. For example, the bioimpedance or tissue dielectric constant data can provide information that correlates with local tissue hydration, or can provide information about different skin layers or cellular structure information. Furthermore, bioimpedance or tissue dielectric constant sensors can provide real-time measurements that can provide information about physiological “noise” within the tissue, which can be used to calibrate other measurements or calculations. As described herein, in some cases, the system can select a focal depth of the Raman spectrometer based at least in part on tissue geometry data.

(95) Raman Spectroscopy

(96) The Raman effect is a light-scattering phenomenon that can provide insight as to one or more characteristics of an analyte in a sample. When light irradiates a tissue, a fraction of the light is scattered, meaning it emerges in directions other than that of the incident (incoming) beam. Most of this scattered light, generally referred to as Rayleigh scattering, emerges at the original frequency (f_0) and wavelength of the incident beam. A small portion of the scattered light, however, emerges at some shifted frequency (f_s) that is different from, and usually lower than, the original frequency (f_0) and has wavelengths different from that of the incident light. The process leading to this small portion of the scattered light is termed the Raman effect or Raman scattering.

(97) Raman scattering can occur with a change in vibrational or rotational energy of a molecule. Accordingly, the Raman spectra can contain information about the specific chemical substance in the irradiated tissue. For example, Raman scattering yields a set of characteristic peaks in a spectrum, which is a “fingerprint” of a specific chemical substance. Therefore, Raman has high specificity in glucose measurements.

(98) Raman spectroscopy has exhibited promise with respect to blood glucose detection, for example, due to its capability to gain information about the molecular constitution non-invasively. For example, features (such as peaks) of the Raman spectra are considered the Raman “fingerprints” of analytes, such as glucose. Accordingly, using an isolated or semi-isolated Raman signal, the system can identify physiological data, such as information regarding a patient's blood glucose level. However, for various reasons, it has been challenging to isolate a pure Raman signal from a signal obtained from a Raman spectrometer.

(99) The signal collected through Raman spectroscopy is based at least in part on the collection optics and the focal distance/depth of the optics into the tissue. In some cases, the system can use data from one or more sensors to select an appropriate focal depth. For example, a focal depth can be selected that may provide a high or the highest resolution of the Raman or collected signal. In addition or alternatively, a focal depth can be selected that will allow the Raman spectrometer to focus on a particular location of the tissue, such as the capillary beds. For example, OCT, bioelectrical impedance, or tissue dielectric constant measurements may provide tissue geometry data (for example, structural and functional information) that can be used to select a focal depth into the tissue. For example, the selection can be based at least in part on a water content of a portion of the tissue, a thickness of one or more skin layers, or a particular location of tissue, such as the capillary beds.

(100) Although complex, an approximation of a measurement obtained from a Raman spectrometer can be determined using one or more of the following equations:

$$(101) \quad I_1 = I_0 e^{-A_1} \quad (\text{Equation3}) \quad R_0 = R_A I_1 \quad (\text{Equation4}) \quad F_0 = I_1 \quad (\text{Equation5})$$

$$I_2 = \text{.Math.}((R_0 + F_0)e^{-A_2}) \quad (\text{Equation6})$$

where $I_{\text{sub}.0}$ is an intensity of excitation light, $I_{\text{sub}.1}$ is an intensity of scattered light, $A_{\text{sub}.1}$ is a first interrogation volume, $R_{\text{sub}.A}$ represents Raman activity, $R_{\text{sub}.0}$ is an intensity Raman scattering at a specific wavelength of light, $F_{\text{sub}.0}$ is an intensity of Florescence at the specific wavelength of light, Φ represents quantum efficiency, $A_{\text{sub}.2}$ represents a second interrogation volume, and $I_{\text{sub}.2}$ is an intensity of measured light. From these relationships, it can be seen that the intensity of measured light ($I_{\text{sub}.2}$) is dependent on the intensity of Raman scattering

(R.sub.0), the intensity of Fluorescence (F.sub.0), or the second interrogation volume (A.sub.2), among other things. Due to the nature of the Raman spectroscopy, the intensity of Raman scattering (R.sub.0) is often of very low intensity. In various aspects, the system can reduce or remove an effect of Fluorescence or absorption on the measured signal, thereby isolating or improving the Raman signal (R.sub.0).

Fluorescence

(102) A challenge in the implementation of Raman spectroscopy to obtain physiological data is the emission of fluorescence. Accordingly, if fluorescence is generated, it often overwhelms the Raman signal, effectively hiding the Raman features. Thus, in some cases, it can be advantageous to isolate the Raman signal.

(103) FIG. 6 shows a graph 600 illustrating various example light intensity signals acquired at a patient's wrist. In this example, the y-axis corresponds to arbitrary intensity units, while the x-axis corresponds to a wavenumber shift (in cm^{-1}). Because the Raman signal is dependent on the excitation wavelength, it can be convenient to use wavenumber to indicate the change of wavelength compared to excitation wavelength. Wavelength change is also photo energy change that is often described by wavenumber change in the frequency domain, because wavenumber is used to describe wavelength in the frequency domain. Wavelength can convert to wavenumbers by dividing one centimeter by wavelength.

(104) As described herein, the light intensity signal acquired from a Raman spectrometer is influenced by the emission of fluorescence. For example, fluorescence is often much more intense than Raman scattering, and fluorescence can overwhelm or mask a Raman measurement in the light intensity signal. This can be seen in each of the signals of the graph 600. For example, the overall shape of each signal of the graph 600 is attributable to the fluorescence, while the subtle oscillations of each signal are attributable to Raman. Because the fluorescence tends to mask the Raman spectrum, it can be desirable to remove or reduce an effect of the fluorescence on the light intensity signal.

(105) Various techniques for removing or reducing an effect of the fluorescence on the light intensity signal are known, including, but not limited to, confocal configuration, photobleaching, chemical bleaching, deployment of laser excitation at longer wavelengths, filtering with respect to pixel frequency (or wavenumber frequency), signal decomposition by various forms of component subtraction from a priori information, photobleaching curve fitting to subtract away an approximated fluorescence signal, frequency offset Raman methods, spatial offset Raman methods, or the like.

(106) For example, irradiating tissue with intense laser light for a long period of time (sometimes referred to as photobleaching) can reduce a level of fluorescence emission in the light intensity signal, thus increasing the signal to noise (S/N) ratio of a Raman measurement. That is because the fluorescence signal of skin will decrease over time (experiencing an exponential decay) as a source is continually shining, while a Raman signal will not change. By looking at the exponential decay (in time) of photobleaching, the system can obtain a fluorescence approximation by curve fitting.

(107) As another example, a system can use a first excitation wavelength to characterize the fluorescence, and then can subtract the fluorescence from a signal of a second excitation wavelength to isolate the Raman. For example, a location of peaks of the fluorescence emission are independent of excitation wavelength, whereas a location of peaks and compactness of emission of Raman spectra are dependent on excitation wavelength. Using this information, the system can remove or reduce an effect of fluorescence emission in the light intensity signal. Fluorescence can also be removed by taking sequential measurements of the tissue over time. For example, the fluorescence signal can be isolated by the change of the measured spectrum overtime.

(108) FIG. 7 illustrates a scaled view of the various example light intensity signals of FIG. 6. As described herein with respect to FIG. 6, the light intensity signals are influenced by, among other things, fluorescence, Raman scattering, and tissue absorption. For example, the light intensity

signals can include a significant fluorescence baseline.

(109) FIG. 8 illustrates an approximation of an intensity of the fluorescence portion **800** of the light intensity signals **700** of FIG. 7. This approximation of fluorescence can be determined using various techniques, such as those described herein. The system can utilize photobleaching curve fitting to subtract away an approximated fluorescence signal. For example, over time, the Raman signal (R.sub.0) will remain constant while the fluorescence F.sub.0 will experience an exponential decay. By looking at the exponential decay (in time) of photobleaching, the system can obtain a fluorescence approximation by curve fitting.

(110) FIG. 9 illustrates an approximation of an intensity of the isolated Raman with tissue absorption signals of FIG. 7. In this example, at least some of the effect of fluorescence (for example, illustrated in FIG. 8) has been removed or reduced. Accordingly, the graph **700** of FIG. 7 can be approximately equal to the Raman and tissue absorption portion (for example, the $\Sigma(R_{sub.0e} - A_{sup.2})$ portion of Equation 6) of the light intensity signals of FIG. 6. As can be seen from a comparison of FIGS. 7 and 9, the presence of fluorescence in the light intensity signals **700** can mask many of the Raman features, such as the peaks, valleys, amplitude, compaction, and the like. By removing or reducing the presence of fluorescence in the light intensity signals **700**, the system can isolate the Raman signal.

(111) FIG. 10 illustrates an approximation of an intensity of the isolated Raman with tissue absorption signals of FIG. 7. In this example, the signal of graph **900** of FIG. 9 has been filtered to reduce or remove at least some of a remaining effect of fluorescence. For example, the system can filter the signal using a band pass or high pass filter.

(112) Absorption

(113) Another challenge in the implementation of Raman spectroscopy to obtain physiological data is the attenuation of the signal due to absorption. In some cases, the Raman signal can be isolated or improved by reducing or removing an effect of absorption on the measured signal. For example, sensor data from one or more sensors, such as a near infrared (NIR), reflectance, transmittance, or pulse oximetry sensor, can be utilized to determine absorption, which can be removed from one or more other measurements, such as a Raman measurement.

(114) An effect of the tissue absorption (for example, the e.sub.-A portion of Equation 6) may be removed or reduced in various ways. For example, the absorption data, transmission data, reflectance data, or the like may be determined using data from one or more sensors, such as, but not limited to, a near infrared (NIR), reflectance, transmittance, or pulse oximetry sensor. Based on the sensor data, a processor can further process the signal (for example, signal **900** or **1000**) to reduce or subtract an effect of the attenuation of the signal due to absorption.

(115) Tissue Geometry

(116) Tissue geometry can vary greatly between individuals. For example, skin structure or skin thickness can vary across races, ages, or the like. Even individuals having similar demographics can have different skin geometries. FIGS. 11A-11C illustrate optical scattering differences in skin geometries among various age groups. FIG. 11A corresponds to 20-39 year olds, FIG. 11B corresponds to 40-59 year-olds, and FIG. 11C corresponds to 60-79 year-olds. In these examples, the x-axis corresponds to a compaction of the skin and is measured from 0 to 200 units, where one unit is 3 μm , and the y-axis corresponds to brightness (for example, backscattered intensity) of the skin and is measured from 0 to 800 AU (absorbance units). As evidenced by these graphs **1100A**, **1100B**, **1100C**, the general skin structure or thickness is not constant throughout the population.

(117) Tissue geometry can also vary greatly between tissue sites of a particular individual. For example, each of a finger, a thumb, a thenar space of a hand, a wrist, a forearm, a nose, an ear, a neck, or other tissue site can have a different skin geometry. Even tissue sites that are in close proximity, such as an upper part of a finger and a lower part of a finger, can have a different skin geometry.

(118) Example Sensor Fusion Apparatus

(119) A patient monitoring system such as systems **100** or **200** can include multiple noninvasive sensors. At least one sensor can be configured to provide tissue geometry information, and the system can utilize tissue geometry data to calibrate one or more other sensors or otherwise improve data obtained by the one or more other sensors. Techniques for utilizing sensor data to improve or calibrate another sensor can be referred to as data harmonization or sensor fusion.

(120) As described herein, data acquired by a sensor can be a function of, or at least affected by, the tissue geometry of the particular tissue site that the sensor is interrogating. For example, tissues having a different geometry can result in a different optical profile. Consequently, data obtained from a first sensor at a first tissue site might not be useful for calibrating or improving a sensor that is interrogating a second, different tissue site. Accordingly, to accurately or reliably harmonize data between sensors, it can be helpful for each of the multiple sensors to acquire data associated with the same or a similar tissue site. In other words, it can be advantageous for each of the multiple sensors to interrogate the same or a sufficiently proximate tissue site so that a variable or other information determined using data from one sensor can be used to improve one or more others sensors. The present disclosure can provide for an apparatus configured allow multiple sensors to interrogate the same or a sufficiently proximate tissue site.

(121) FIGS. **12A-12B** illustrate an example sensor fusion apparatus **1200** configured with multiple sensing capabilities for interrogation of the same or a sufficiently proximate tissue site. As shown, the apparatus **1200** can include an OCT sensor or lens **1202**, a Raman spectrometer **1204**, a pulse oximetry sensor **1206**, and a bioimpedance sensor **1208**.

(122) As illustrated in FIG. **12B**, the apparatus **1200** can include a cylindrical housing **1210**. In the illustrated example, the sensor side of the apparatus **1200** can be positioned on or proximate to a tissue site of a patient, and one or more of the an OCT sensor **1202**, a Raman spectrometer **1204**, a pulse oximetry sensor **1206**, and a bioimpedance sensor **1208** can be configured to interrogate the same or a sufficiently proximate tissue site. Although at least some of the sensors are illustrated as being configured to obtain data via reflectance technologies, in some cases one or more sensors are configured to obtain data via transmittance or other technologies.

(123) Example Reflectance Sensor

(124) FIG. **13** illustrates an example reflectance plethysmography sensor or probe **1300**. The reflectance sensor or probe **1300** includes a light source **1302** at its center and seven detector channels **1304** surrounding the light source **1302**. The light source **1302** and emit light to illuminate a tissue site, and one or more of the channels **1304** can detect the light after it interacts with the tissue site. In some cases, the one or more of the channels **1304** can generate a composite analog light intensity signal responsive to the detected light. In some cases, light source **1302** or the channels **1304** can include a fiber-optic component for illumination or collection. For example, the light source **1302** can include a fiber bundle.

(125) FIG. **14** illustrates an environment **1400** that shows a hand of a user interacting with the example reflectance sensor **1300** of FIG. **13**. The reflectance sensor **1300** can be configured to interact with one or more of the various tissue sites described herein. For example, as illustrated, the reflectance sensor **1300** can be positioned to interrogate at a metacarpal bone **1306**. The metacarpal bone **1306** forms the intermediate part of the skeletal hand located between the phalanges of the fingers and the carpal bones of the wrist which forms the connection to the forearm.

(126) Example Patient Monitoring

(127) FIG. **15** illustrates an example physiological monitoring system **1500**, which can be an embodiment of the patient monitoring system **100** or **200**. As illustrated in FIG. **15**, the system **1500** can include a first sensor **1504A** and a second sensor **1504B**. In some implementations, the first sensor **1504A**, the second sensor **1504B**, or another sensor can be integrated into an apparatus, such as a wearable apparatus like a glove, a sock, an armband, a headband, a chest strap, etc.

(128) The first sensor **1504A** can be similar to sensor **204A**, as described herein with respect to

FIG. 2. For example, the sensor **1504A** can include an emitter and detector. The emitter can emit light (for example, of an infrared or near-infrared wavelength) to illuminate a tissue site of a patient. In this example, the tissue site corresponds to a thenar space of the patient's hand. However, other tissue sites are contemplated. As the light interacts with (for example, passes through) the thenar space of the hand, some light may be absorbed, reflected, refracted, or the like. The detector can receive or generate a signal responsive to the light detected by the detector after it interacts with the thenar space of the hand. The signal generated by the detector can be received by a processor (not shown), which can determine one or more various physiological parameters, such as an absorbance of the tissue based at least in part on the received signal.

(129) The second sensor **1504B** can be similar to sensor **204C**, as described herein with respect to FIG. 2. For example, the second sensor **1504B** can include a light source and a detector. The emitter can emit light (for example, of an infrared or near-infrared wavelength) to illuminate a tissue site of a patient. As the light interacts with (for example, reflects off) the thenar space of the hand, some light may be absorbed, transmitted through, reflected, refracted, or the like. The detector can receive or generate a signal responsive to the light detected by the detector after it interacts with the hand. The signal generated by the detector can be received by a processor (not shown), which can determine one or more various physiological parameters, such as a transmittance of the tissue based at least in part on the received signal.

(130) FIGS. **16A**, **16B**, and **16C** illustrate an example physiological monitoring apparatus **1600**. As illustrated, a user can place his or her arm in the apparatus **1600**, such that two or more sensors of the apparatus **1600** can interrogate tissue of the arm. The two or more sensors can correspond to any of the sensors described herein. For example, the two or more sensors can interrogate the same or a different tissue site of the arm. In some cases, the apparatus **1600** can be miniaturized and integrated into a wearable apparatus, such as a glove, a sock, an armband, a headband, a chest strap, etc.

(131) Example Data Harmonization

(132) FIG. **17** illustrates a flow diagram illustrative of an example routine for harmonizing data from a plurality of non-invasive sensors. One skilled in the relevant art will appreciate that the elements outlined for routine **1700** may be implemented by one or many computing devices/components, such as in hardware, with a front end component, with a sensor interface, or with a processor, such as one or more processors housed in a patient monitor, one or more remote processors, one or more processors housed in the sensors, etc. Accordingly, although routine **1700** has been logically associated as being generally performed by a processor, the following illustrative embodiments should not be construed as limiting.

(133) At block **1702**, a processor can receive data from one or more first noninvasive sensors. The one or more first noninvasive sensors can include an optical coherence tomography (OCT) sensor. As described herein, the OCT sensor can provide a non-invasive method for identifying one or more characteristics of a tissue's structure. The data received by the processor from the OCT sensor can include OCT data, which can be referred to as tissue geometry data.

(134) In addition or alternatively, the one or more first noninvasive sensors can include a bioimpedance sensor or a tissue dielectric constant sensor. As described herein, the bioimpedance sensor or tissue dielectric constant sensor can provide a non-invasive method for identifying one or more characteristics of a tissue's structure. The data received by the processor from the bioimpedance sensor or tissue dielectric constant sensor can include bioimpedance data, which can include tissue geometry data, hydration data, or the like.

(135) At block **1704**, a processor can receive data from one or more second noninvasive sensors. The one or more second noninvasive sensors can include a pulse oximetry sensor, such as a reflectance or transmission sensor. As described herein, the pulse oximetry sensor can provide a non-invasive method for identifying one or more of various physiological parameters.

(136) At block **1706**, a processor can receive data from one or more third noninvasive sensors. The

one or more second noninvasive sensors can include a Raman spectrometer. As described herein, the Raman spectrometer can provide a non-invasive method for identifying or more of various physiological parameters.

(137) At block **1708**, the processor can harmonize the data received from two or more of the non-invasive sensors. By harmonizing the data from two or more non-invasive sensors, the system may be able to compensate for circumstances that might otherwise result in inaccurate or unreliable data. For example, using skin geometry information (for example, skin thickness), the processor can weight or prioritize longer or shorter path length detectors. In addition or alternatively, the various sensor data, such as skin geometry information, can allow the processor compensate for sensor or probe placement. For example, a location, coupling, or pressure can be compensated by the processor by adjusting path length, which can be determined from the various sensor data, such as skin geometry information. Similarly, the processor can utilize the various sensor data, such as skin geometry information, to detect drift or motion at the tissue site.

(138) As a non-limiting example, the data received at block **1702** from the OCT sensor, the bioelectrical impedance sensor, or the tissue dielectric constant sensor can include tissue geometry information. Based at least in part on the tissue geometry data, the processor can determine a path length corresponding to a tissue site interrogated by the one or more first noninvasive sensors. In some cases, the determined path length can be utilized with the pulse oximetry sensor to determine a concentration of an analyte, such as blood glucose. For example, based on the data received at block **1704** from the one or more second noninvasive sensors, the processor can determine an absorbance corresponding to a tissue site interrogated by the one or more second noninvasive sensors. Using one or more relationships derived from Beer's law (Equation 1), the concentration, c , of one or more analytes can be determined using the absorbance, A , determined from the pulse oximetry sensor data, and the path length, b , determined from the tissue geometry data.

(139) As another non-limiting example, the processor can utilize the tissue geometry data to select a focal depth or focal length, wavelength, refractive index, or other parameter associated with the Raman spectrometer. For example, the tissue geometry data can provide an indication of a particular location of tissue, such as the capillary beds. The processor can select a focal depth or focal length of the Raman spectrometer such that the Raman spectrometer can focus on this particular location. As a result, the processor can determine a more accurate indication of glucose concentration from the Raman signal.

(140) As another non-limiting example, the processor can utilize the pulse oximetry data to filter data received from a Raman Spectrometer to isolate a Raman Spectra. For example, as described herein, a direct measurement of glucose can be determined based on features of the isolated Raman signal. Using the pulse oximetry data, the processor can filter out an effect of absorbance on the Raman Spectra.

(141) In addition or alternatively, using the various sensor data, the processor can create calibrations for one or more individuals. For example, although skin geometry may vary between individuals, one or more groups of individuals may have similar skin geometries, which can allow for more accurate physiological parameter estimations of for individuals in those groups. For example, using the various sensor data, such as the skin geometry, Raman, or NIR data, the processor can determine calibrations for different groups, such as different skin populations, different ages, or the like.

(142) The various blocks of process **1700** described herein can be implemented in a variety of orders, and that the system can implement one or more of the blocks concurrently or change the order, as desired. For example, the system **100** can concurrently receive any of the sensor data, or receive the sensor data in any order. Similarly, the system can make one or more calculations or determinations in any order, such as before or after receiving data from one or more sensors.

(143) It will be understood that any of the first, second, or third sensors can interrogate the same or a different tissue site. Furthermore, it will be understood that fewer, more, or different blocks can

be used as part of the routine **1700**. Likewise, fewer, more, or different sensors can be used by the system. For example, the routine **1700** can include blocks for receiving data associated with additional non-invasive sensors or determining various other physiological parameters. Furthermore, the routine **1700** can include causing a display to display one or more of various indications of the any other the sensor data, calculations, or determinations.

(144) FIG. **18** illustrates a flow diagram illustrative of an example routine for harmonizing data from a plurality of non-invasive sensors. One skilled in the relevant art will appreciate that the elements outlined for routine **1800** may be implemented by one or many computing devices/components, such as in hardware, with a front end component, with a sensor interface, or with a processor, such as one or more processors housed in a patient monitor, one or more remote processors, one or more processors housed in the sensors, etc. Accordingly, although routine **1800** has been logically associated as being generally performed by a processor, the following illustrative embodiments should not be construed as limiting.

(145) At block **1802**, the process **1800** can receive tissue geometry data from a first noninvasive sensor. As described herein, the first non-invasive sensor can include a combination of one or more of an OCT sensor, a bioimpedance sensor, a tissue dielectric constant sensor, or any other sensor configured to measure or determine tissue geometry data. The tissue geometry data can include various information corresponding to the skin, fluids, bones, or the like. For example, tissue geometry data can include, but is not limited to, a thickness of one or more skin layers (for example, the epidermis, the dermoepidermal junction, the papillary dermis, the reticular dermis, etc.), cellular structure information, a water content of a portion of the tissue, etc.

(146) At block **1804**, the process **1800** can calibrate a Raman Spectrometer based at least in part on the tissue geometry data received at block **1802**. For example, the tissue geometry data can provide insight about the tissue site, which can allow the process **1800** to optimize one or more settings of the Raman spectrometer. For example, based at least in part on the tissue geometry data, the process **1800** can select a focal depth or focal length, wavelength, refractive index, or other parameter associated with the Raman spectrometer. By adjusting one or more settings or positioning of the Raman spectrometer based on the tissue geometry data, the process can enhance a signal received by the Raman spectrometer. For example, the new settings can increase the collection efficiency, the resolution, the signal-to-noise ratio, or the like of the Raman signal.

(147) At block **1806**, the process **1800** can receive absorption, transmission, reflectance, or other data from a second noninvasive sensor. As described herein, the second non-invasive sensor can include one or more of a pulse oximetry sensor, a reflectance sensor, a transmittance sensor, or another sensor from which absorption, transmission, reflectance, or other tissue related data can be determined. In some cases, the second noninvasive sensor can include a light source configured to emit light and a detector and configured to detect light. Depending on the type of sensors, the detected can be configured to detect light after having it has passed through, reflected, refracted, or scattered at a tissue site of a patient. In some cases, the tissue site corresponding to the second sensor (for example, the tissue site at which the second sensor takes a measurement) is the same tissue site (or within a close proximity) as the tissue site of the second sensor. For example, the first and second sensors can be configured to interrogate the tissue site at different periods of time. However, in some cases, the first and second sensors can be configured to interrogate different tissue sites.

(148) At **1808**, the process **1800** can receive a Raman signal corresponding to the Raman spectrometer. As described herein, the light intensity signal acquired from a Raman spectrometer is influenced by the emission of fluorescence.

(149) At block **1810**, the process **1800** can determine an isolated Raman signal by reducing or removing an effect of fluorescence or an effect of absorption from the Raman signal received at block **1808**. As described herein, fluorescence can overwhelm or mask a Raman measurement in the light intensity signal. As such, the process **1800** can use one or more techniques described

herein to reduce or remove an effect of the fluorescence on the Raman signal. In addition or alternatively, the process **1800** can reduce or remove an effect of absorption on the Raman signal. For example, using the absorption data acquired at block **1806**, the process **1800** can filter, subtract, reduce, or remove an effect of absorption on the Raman signal. By reducing or removing an effect of fluorescence or an effect of absorption from the Raman signal, the process **1800** can determine an isolated (or semi-isolated) Raman signal.

(150) At block **1812**, the process **1800** can determine data corresponding to one or more analytes based at least in part on the isolated Raman signal. For example, features of the Raman spectra (such as peaks, valleys, concentrations, etc.) can correspond to analytes such as glucose. Accordingly, using the isolated or semi-isolated Raman signal, the system can identify physiological data, such as information regarding a patient's blood glucose level. Thus, the process **1800** can harmonize data from various non-invasive sensors to non-invasively determine a patient's blood glucose level, or other analyte.

(151) It will be understood that the various blocks of process **1800** described herein can be implemented in a variety of orders, and that the system can implement one or more of the blocks concurrently or change the order, as desired. Furthermore, it will be understood that fewer, more, or different blocks can be used as part of the routine **1800**. For example, fewer, more, or different sensors can be used by the system. Furthermore, the routine **1800** can include blocks for receiving data associated with additional non-invasive sensors or determining various other physiological parameters. Furthermore, the routine **1800** can include displaying one or more of various indications of the any other the sensor data, calculations, or determinations.

Further Examples

(152) Various example features can be found in the following clauses, which can be implemented together with any combination of the features described above:

(153) Clause 1: A physiological monitoring system configured to determine a physiological parameter by harmonizing data between two or more different types of non-invasive physiological sensors interrogating the same or proximate measurement sites, the physiological monitoring system comprising: a first non-invasive sensing device of a first type configured to interrogate a tissue site of a patient and generate a first signal indicative of first physiological data associated with the tissue site; a second non-invasive sensing device of a second type that is different from the first type configured to interrogate the tissue site of the patient and generate a second signal indicative of second physiological data associated with the tissue site; and one or more processors in communication with the first and second non-invasive sensing devices, the one or more processors configured to: receive the first signal indicative of the first physiological data; receive the second signal indicative of the second physiological data; and determine a physiological parameter based at least in part on the first and second signals.

(154) Clause 2: The system of Clause 1, wherein the first non-invasive sensing device comprises one or more of an optical coherence tomography (OCT) sensor, a bioimpedance sensor, or a tissue dielectric constant sensor.

(155) Clause 3: The system of any of the previous clauses, wherein the first physiological data comprises tissue geometry data corresponding to the tissue site.

(156) Clause 4: The system of any of the previous clauses, wherein tissue geometry data comprises at least one of data corresponding to one or more of a thickness of one or more layers of skin of the tissue site, cellular structure information associated with the tissue site, or a water content associated with the tissue site.

(157) Clause 5: The system of any of the previous clauses, wherein the first non-invasive sensing device comprises a plethysmography sensor.

(158) Clause 6: The system of any of the previous clauses, wherein the plethysmography sensor comprises one or more of a pulse oximetry sensor, a transmission plethysmography sensor, or a reflectance plethysmography sensor.

(159) Clause 7: The system of any of the previous clauses, wherein the plethysmography sensor comprises: at least one emitter configured to emit light, and at least one detector configured to: detect the light after interaction with the tissue site, and generate the second signal responsive to the detected light.

(160) Clause 8: The system of any of the previous clauses, wherein the first non-invasive sensing device and the second non-invasive sensing device are configured to simultaneously interrogate the tissue site.

(161) Clause 9: The system of any of the previous clauses, wherein the first non-invasive sensing device and the second non-invasive sensing device are configured to interrogate the tissue site at distinct and different time periods.

(162) Clause 10: The system of any of the previous clauses, wherein the second non-invasive sensing device comprises a Raman spectrometer.

(163) Clause 11: The system of any of the previous clauses, wherein the second physiological data corresponds to Raman spectra associated with the tissue site.

(164) Clause 12: The system of any of the previous clauses, wherein the one or more processors are further configured to determine an isolated Raman signal by reducing or removing at least one of an effect of fluorescence or an effect of absorption from the second signal.

(165) Clause 13: The system of any of the previous clauses, wherein the one or more processors are further configured to determine the isolated Raman signal based at least in part on the first signal.

(166) Clause 14: The system of any of the previous clauses, wherein the one or more processors are further configured to determine a path length associated with the tissue site based at least in part on the first signal, wherein the determination of the physiological parameter is based at least in part on the determined path length.

(167) Clause 15: The system of any of the previous clauses, wherein the one or more processors are further configured to determine an absorption of light of the tissue site based at least in part on the second signal, wherein the determination of the physiological parameter is based at least in part on the determined absorption of light.

(168) Clause 16: The system of any of the previous clauses, wherein physiological parameter comprises a concentration of one or more analytes.

(169) Clause 17: The system of any of the previous clauses, wherein physiological parameter comprises a blood glucose concentration associated with the tissue site.

(170) Clause 18: The system of any of the previous clauses, wherein one or more processors are further configured to calibrate the second non-invasive sensing device based at least in part on the first signal.

(171) Clause 19: The system of any of the previous clauses, wherein second non-invasive sensing device comprises a Raman spectrometer, wherein to calibrate the Raman spectrometer, the one or more processors are configured to select at least one of a focal depth, a focal length, a wavelength, or a refractive index, associated with the Raman spectrometer.

(172) Clause 20: The system of any of the previous clauses, wherein the tissue site comprises a thenar space of a hand.

(173) Clause 21: The system of any of the previous clauses, wherein the tissue site comprises an area associated with a metacarpal bone.

(174) Clause 22: A method for determining a physiological parameter by harmonizing data between two or more different types of non-invasive physiological sensors interrogating the same or proximate measurement sites, the method comprising: receiving a first signal from a first non-invasive sensing device of a first type, wherein the first non-invasive sensing device is configured to interrogate a tissue site of a patient and generate the first signal, wherein the first signal is indicative of first physiological data associated with the tissue site; receiving a second signal from a second non-invasive sensing device of a second type that is different from the first type, wherein the second non-invasive sensing device is configured to interrogate the tissue site of the patient and

generate the second signal, wherein the second signal is indicative of second physiological data associated with the tissue site; and/or determining a physiological parameter based at least in part on the first and second signals.

(175) Clause 23: The method of Clause 22, wherein the first non-invasive sensing device comprises one or more of an optical coherence tomography (OCT) sensor, a bioimpedance sensor, or a tissue dielectric constant sensor.

(176) Clause 24: The method of any of Clauses 22 or 23, wherein the first physiological data comprises tissue geometry data corresponding to the tissue site.

(177) Clause 25: The method of any of Clauses 22-24, wherein tissue geometry data comprises at least one of data corresponding to one or more of a thickness of one or more layers of skin of the tissue site, cellular structure information associated with the tissue site, or a water content associated with the tissue site.

(178) Clause 26: The method of Clauses 22-25, wherein the first non-invasive sensing device comprises a plethysmography sensor.

(179) Clause 27: The method of Clauses 22-26, wherein the plethysmography sensor comprises one or more of a pulse oximetry sensor, a transmission plethysmography sensor, or a reflectance plethysmography sensor.

(180) Clause 28: The method of Clauses 22-27, wherein the plethysmography sensor comprises: at least one emitter configured to emit light, and at least one detector configured to: detect the light after interaction with the tissue site, and generate the second signal responsive to the detected light.

(181) Clause 29: The method of Clauses 22-28, wherein the first non-invasive sensing device and the second non-invasive sensing device are configured to simultaneously interrogate the tissue site.

(182) Clause 30: The method of Clauses 22-29, wherein the first non-invasive sensing device and the second non-invasive sensing device are configured to interrogate the tissue site at distinct and different time periods.

(183) Clause 31: The method of Clauses 22-30, wherein the second non-invasive sensing device comprises a Raman spectrometer.

(184) Clause 32: The method of Clauses 22-31, wherein the second physiological data corresponds to Raman spectra associated with the tissue site.

(185) Clause 33: The method of Clauses 22-32, further comprising determining an isolated Raman signal by reducing or removing at least one of an effect of fluorescence or an effect of absorption from the second signal.

(186) Clause 34: The method of Clauses 22-33, wherein said determining the isolated Raman signal is based at least in part on the first signal.

(187) Clause 35: The method of Clauses 22-34, further comprising determining a path length associated with the tissue site based at least in part on the first signal, wherein said determining the physiological parameter is based at least in part on the determined path length.

(188) Clause 36: The method of Clauses 22-35, further comprising determining an absorption of light of the tissue site based at least in part on the second signal, wherein said determining the physiological parameter is based at least in part on the determined absorption of light.

(189) Clause 37: The method of Clauses 22-36, wherein physiological parameter comprises a concentration of one or more analytes.

(190) Clause 38: The method of Clauses 22-37, wherein physiological parameter comprises a blood glucose concentration associated with the tissue site.

(191) Clause 39: The method of Clauses 22-38, further comprising calibrating the second non-invasive sensing device based at least in part on the first signal.

(192) Clause 40: The method of Clauses 22-39, wherein second non-invasive sensing device comprises a Raman spectrometer, the method further comprising calibrating the Raman spectrometer by selecting at least one of a focal depth, a focal length, a wavelength, or a refractive index, associated with the Raman spectrometer.

(193) Clause 41: The method of Claims 22-40, wherein the tissue site comprises a thenar space of a hand.

(194) Clause 42: The method of Claims 22-41, wherein the tissue site comprises an area associated with a metacarpal bone.

(195) Clause 43: A physiological monitoring device comprising any of the features of any of the previous claims.

Terminology

(196) The term “and/or” herein has its broadest least limiting meaning which is the disclosure includes A alone, B alone, both A and B together, or A or B alternatively, but does not require both A and B or require one of A or one of B. As used herein, the phrase “at least one of” A, B, “and” C should be construed to mean a logical A or B or C, using a non-exclusive logical or.

(197) The following description is merely illustrative in nature and is in no way intended to limit the disclosure, its application, or uses. For purposes of clarity, the same reference numbers will be used in the drawings to identify similar elements. It should be understood that steps within a method may be executed in different order without altering the principles of the present disclosure.

(198) Features, materials, characteristics, or groups described in conjunction with a particular aspect, embodiment, or example are to be understood to be applicable to any other aspect, embodiment or example described herein unless incompatible therewith. All of the features disclosed in this specification (including any accompanying claims, abstract and drawings), or all of the steps of any method or process so disclosed, may be combined in any combination, except combinations where at least some of such features or steps are mutually exclusive. The protection is not restricted to the details of any foregoing embodiments. The protection extends to any novel one, or any novel combination, of the features disclosed in this specification (including any accompanying claims, abstract and drawings), or to any novel one, or any novel combination, of the steps of any method or process so disclosed.

(199) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of protection. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms. Furthermore, various omissions, substitutions and changes in the form of the methods and systems described herein may be made. Those skilled in the art will appreciate that in some embodiments, the actual steps taken in the processes illustrated or disclosed may differ from those shown in the figures. Depending on the embodiment, certain of the steps described above may be removed, others may be added. For example, the actual steps or order of steps taken in the disclosed processes may differ from those shown in the figures. Depending on the embodiment, certain of the steps described above may be removed, others may be added. For instance, the various components illustrated in the figures may be implemented as software or firmware on a processor, controller, ASIC, FPGA, or dedicated hardware. Hardware components, such as processors, ASICs, FPGAs, and the like, can include logic circuitry. Furthermore, the features and attributes of the specific embodiments disclosed above may be combined in different ways to form additional embodiments, all of which fall within the scope of the present disclosure.

(200) User interface screens illustrated and described herein can include additional or alternative components. These components can include menus, lists, buttons, text boxes, labels, radio buttons, scroll bars, sliders, checkboxes, combo boxes, status bars, dialog boxes, windows, and the like. User interface screens can include additional or alternative information. Components can be arranged, grouped, displayed in any suitable order.

(201) Although the present disclosure includes certain embodiments, examples and applications, it will be understood by those skilled in the art that the present disclosure extends beyond the specifically disclosed embodiments to other alternative embodiments or uses and obvious modifications and equivalents thereof, including embodiments which do not provide all of the features and advantages set forth herein. Accordingly, the scope of the present disclosure is not

intended to be limited by the specific disclosures of preferred embodiments herein, and may be defined by claims as presented herein or as presented in the future.

(202) Conditional language, such as “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements, or steps. Thus, such conditional language is not generally intended to imply that features, elements, or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements, or steps are included or are to be performed in any particular embodiment. The terms “comprising,” “including,” “having,” and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list. Further, the term “each,” as used herein, in addition to having its ordinary meaning, can mean any subset of a set of elements to which the term “each” is applied.

(203) Conjunctive language such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, or Z. Thus, such conjunctive language is not generally intended to imply that certain embodiments require the presence of at least one of X, at least one of Y, and at least one of Z.

(204) Language of degree used herein, such as the terms “approximately,” “about,” “generally,” and “substantially” as used herein represent a value, amount, or characteristic close to the stated value, amount, or characteristic that still performs a desired function or achieves a desired result. For example, the terms “approximately,” “about,” “generally,” and “substantially” may refer to an amount that is within less than 10% of, within less than 5% of, within less than 1% of, within less than 0.1% of, and within less than 0.01% of the stated amount. As another example, in certain embodiments, the terms “generally parallel” and “substantially parallel” refer to a value, amount, or characteristic that departs from exactly parallel by less than or equal to 15 degrees, 10 degrees, 5 degrees, 3 degrees, 1 degree, or 0.1 degree.

(205) The scope of the present disclosure is not intended to be limited by the specific disclosures of preferred embodiments in this section or elsewhere in this specification, and may be defined by claims as presented in this section or elsewhere in this specification or as presented in the future. The language of the claims is to be interpreted broadly based on the language employed in the claims and not limited to the examples described in the present specification or during the prosecution of the application, which examples are to be construed as non-exclusive.

Claims

1. A system for determining a physiological parameter of an individual, the system comprising: a sensor apparatus comprising: a first noninvasive sensor configured to generate first data, wherein the first data is responsive to light reflection, light refraction, light absorption, light attenuation, optical coherence tomography, capacitance, or bioimpedance of a particular region of tissue of the individual, a second noninvasive sensor configured to generate second data, wherein the second data corresponds to a light scattering signature of the particular region of the tissue, and a housing configured to secure the first and second noninvasive sensors; and one or more processors configured to: receive the first data, receive the second data, filter the second data based at least in part on the first data, estimate an analyte level based at least in part on the filtered second data, and output a visual or audible indication of the analyte level.
2. The system of claim 1, wherein the first noninvasive sensor is an optical coherence tomography (OCT) device.

3. The system of claim 1, wherein the first noninvasive sensor is a capacitance sensor.
4. The system of claim 1, wherein the first noninvasive sensor is a bioimpedance sensor.
5. The system of claim 1, wherein the first noninvasive sensor is a plethysmography sensor.
6. The system of claim 1, wherein the sensor apparatus comprises a third noninvasive sensor configured to generate third data responsive to light reflection, light refraction, light absorption, light attenuation, optical coherence tomography, capacitance, or bioimpedance of a particular region of the particular region of tissue, wherein the third data is different from the first data, and wherein the one or more processors is further configured to receive the third data, and wherein the one or more processors is configured to select a focal depth or focal length of the second data based at least in part on the third data.
7. The system of claim 1, wherein the first noninvasive sensor and second noninvasive sensor are configured to concurrently interrogate the particular region of the tissue.
8. The system of claim 1, wherein the first noninvasive sensor and second noninvasive sensor are configured to concurrently interrogate the particular region of the tissue.
9. The system of claim 1, wherein the analyte level comprises a blood analyte level.
10. The system of claim 9, wherein the blood analyte level comprises a glucose level.
11. A method for determining a physiological parameter of an individual, the method comprising: receiving, from a sensor apparatus comprising a first noninvasive sensor and a second noninvasive sensor, first data associated with the first noninvasive sensor, wherein the first data is responsive to light reflection, light refraction, light absorption, light attenuation, optical coherence tomography, capacitance, or bioimpedance of a particular region of tissue of an individual, wherein the second noninvasive sensor comprises a Raman spectrometer; receiving, from the sensor apparatus, second data associated with a second noninvasive sensor, wherein the second data corresponds to a light scattering signature of the particular region of the tissue; filtering the second data based on an effect of the first data on the second data; determining an analyte level based at least in part on the filtered second data; outputting a visual or audible indication of the analyte level; and modifying, based at least in part on the analyte level, a treatment of the individual performed by at least one of a drug administration device, a ventilator, a fluid IV, a fan, or a thermostat.
12. The method of claim 11, comprising: receiving, from the sensor apparatus, third data associated with a third noninvasive sensor, wherein the third data is responsive to light reflection, light refraction, light absorption, light attenuation, optical coherence tomography, capacitance, or bioimpedance of a particular region of the particular region of tissue, wherein the third data is different from the first data; and selecting a focal depth or focal length of the second data based at least in part on the third data.
13. The method of claim 11, comprising interrogating, using the first and second noninvasive sensors, the particular region of tissue.
14. The method of claim 13, wherein the interrogating using the first and second noninvasive sensors is simultaneous.
15. The method of claim 11, wherein the first noninvasive sensor is an optical coherence tomography (OCT) device.
16. The method of claim 11, wherein the first noninvasive sensor is a capacitance sensor.
17. The method of claim 11, wherein the first noninvasive sensor is a bioimpedance sensor.
18. The method of claim 11, wherein the first noninvasive sensor is a plethysmography sensor.
19. A system for determining a physiological parameter of an individual, the system comprising: a sensor apparatus comprising: at least one noninvasive sensor configured to generate first data, wherein the first data is responsive to light reflection, light refraction, light absorption, light attenuation, optical coherence tomography, capacitance, or bioimpedance of a particular region of tissue of the individual; and a housing configured to secure the at least one noninvasive sensor; and one or more processors configured to: receive the first data, receive second data responsive to Raman spectra of the particular region of tissue of the individual, filter the Raman spectra based at

least in part on the first data, estimate an analyte level based at least in part on the filtered Raman spectra, and output a visual or audible indication of the analyte level.

20. The system of claim 19, wherein the sensor apparatus comprises a noninvasive sensor configured to generate third data responsive to light reflection, light refraction, light absorption, light attenuation, optical coherence tomography, capacitance, or bioimpedance of a particular region of the particular region of tissue, wherein the third data is different from the first data, and wherein the one or more processors is further configured to receive the third data, and wherein the one or more processors is configured to select a focal depth or focal length of the Raman spectra based at least in part on the third data.
